SATA 1D: Three Open Points

Sub-Band vs Whole Band, Heatmaps at Larger Distances, and Why the Ambiguities Saturate

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Setup. SATA 1D, $N_{rx} = 4$, PRF = 2000 Hz, $\text{PRF}_{\text{ch}} = 500$ Hz, DPCA along-track baselines ($\Delta b_{at} = 7.69$ m), $b_{xt} = 63.7, 27.0, 4.1, 1.7$ m, $r_0 = 766.2$ km, $\theta = 20°$, $\lambda = 0.25$ m, processed band $B_a = 1235$ Hz. Iso-range point targets. Reconstructions: no SATA, SATA whole band, SATA per sub-band.

Point	Answer
1. Sub-band = whole band?	Expected, not a bug. The residual ΔC_0 of a target is the same in every sub-band (to 10^{-6} mm), and both kernels apply the same phase to every time–frequency cell that holds energy. The look-angle-dependent part of the residual is below 0.16° even for 10 km heights. Per sub-band can only differ where the residual changes across the Doppler band (2D range migration, non-parallel tracks, squint).
2. Heatmaps	The reference was the ideal signal, not limited to B_a; near other targets the windows read the main-lobe skirts (−43 to −58 dB) for every method, which capped the improvement. New metric: $\lvert s - s_{flat}\rvert$, with s_{flat} the same reconstruction without topography. With it, separated targets reach −49 to −102 dB after SATA (improvement 32–55 dB). Added spacings of 5.2 and 10 km and a spacing sweep up to 12.5 km (456 + 57 cases).
3. Saturation	The reconstruction redistributes each target's energy among the Doppler bands with a unitary (circulant) matrix set by the channel phases. The ambiguity cannot exceed $\beta_1 = -4.5$ dB, and once the heights exceed one phase cycle of the long baselines (1.1 and 2.6 km) the phases are random and more height changes nothing. The theory predicts the simulation within 0.3 dB. After SATA the plateau (−13 to −14 dB for close targets with large height steps) comes from the two long-baseline channels losing coherence while the two short ones do not.

1 Why SATA per sub-band gives the same result as SATA whole band

Evidence 1: the residual does not depend on the sub-band. The per-sub-band map fits ΔC_0 only over the look angles of sub-band k (`getcoeff_beam`). For a target at $h = 240$ m the four sub-band values equal the whole-aperture value to 1.3×10^{-6} mm on every channel. The reason is the definition: C_0 is the constant of the parabola fitted to the bistatic-minus-monostatic range history, completed to its stationary point (closest approach). It is a property of the target, not of the look angle. Any look-angle dependence would show up in ΔC_1 (linear in Doppler) and ΔC_2 (quadratic). Table 1 gives the phase these terms add across the whole processed band. It stays below 0.16° up to 10 km heights, while ΔC_0 itself reaches thousands of degrees.

h [m]	ch 0 ($b_{xt} = 63.7$ m)	ch 1 (27.0 m)	ch 2 (4.1 m)	ch 3 (1.7 m)
100	$-32.9°$ / $0.002°$	$-13.9°$ / $0.001°$	$-2.1°$ / $0.000°$	$-0.9°$ / $0.000°$
500	$-164.0°$ / $0.008°$	$-69.5°$ / $0.004°$	$-10.5°$ / $0.001°$	$-4.3°$ / $0.000°$
1000	$-327.0°$ / $0.016°$	$-138.5°$ / $0.007°$	$-21.0°$ / $0.001°$	$-8.5°$ / $0.000°$
3000	$-969.7°$ / $0.049°$	$-410.7°$ / $0.021°$	$-62.4°$ / $0.003°$	$-25.2°$ / $0.001°$
10000	$-3112.0°$ / $0.157°$	$-1318.1°$ / $0.066°$	$-200.2°$ / $0.010°$	$-80.7°$ / $0.004°$

Table 1: Per channel: residual phase $2\pi\Delta C_0/\lambda$ / peak-to-peak variation of the residual phase across $|f| \le B_a/2$, i.e. of $2\pi[(\Delta D_t - \Delta C_1)f + \Delta C_2 \lambda f^2]$. A per-sub-band correction could at most follow this variation.

Evidence 2: both kernels apply the same phase to every cell with energy. A channel line is sampled at $\mathrm{PRF}_{\mathrm{ch}}$, so in each sub-aperture spectrum a target occupies one bin, at its Doppler modulo $\mathrm{PRF}_{\mathrm{ch}}$. The whole-band kernel labels that bin with the baseband frequency, so the fold m of the target lands $m \cdot 6.23$ km away from it; the map carries images of every block at $\{-X, 0, +X\}$ for this reason. The kernel of sub-band k labels the same bin with a frequency in $[f_k - \mathrm{PRF}_{\mathrm{ch}}/2, f_k + \mathrm{PRF}_{\mathrm{ch}}/2)$, so the same fold lands at a different image of the *same* target, with the same value. The occupied cells get the same correction (Fig. 1): identical for one target, and different on 0.011 % of the energy for two targets 2.6 km apart, where the image sets of the two kernels overlap differently.

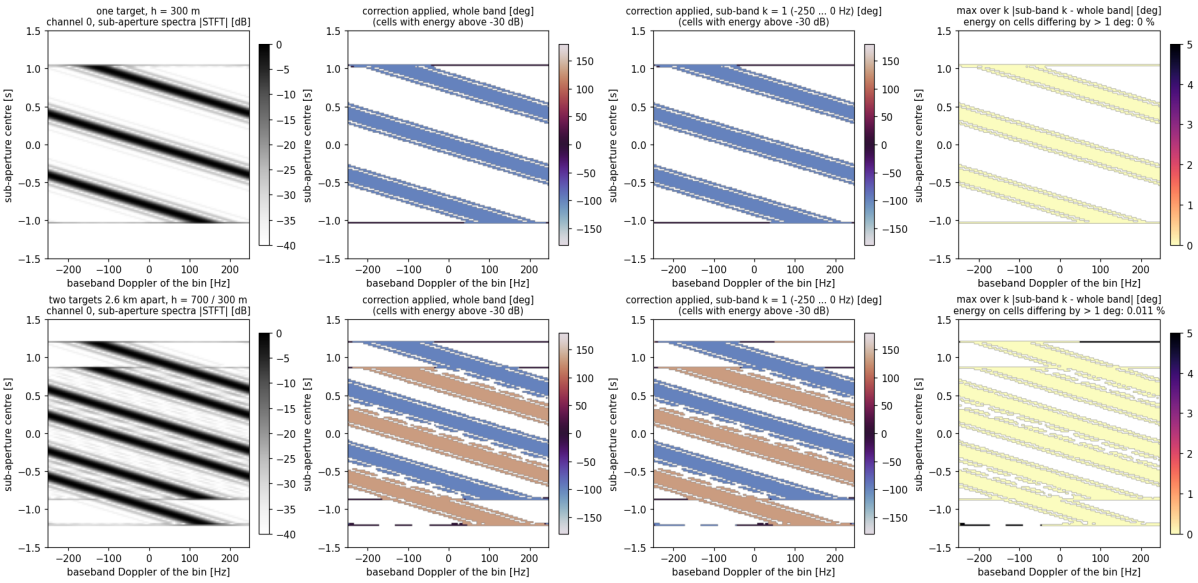


Figure 1: Channel 0, sub-aperture spectra of the channel line (left), phase applied by the whole-band kernel and by the kernel of sub-band 1, and the largest difference over the four sub-band kernels, on the cells within 30 dB of the peak.

Evidence 3: the reconstructions agree band by band. The four corrected channel sets differ from the whole-band set by 0.00–0.71 %, and the error of every output sub-band is the same for both schemes (Fig. 2). Over the 456 cases of the sweep (Section 2), sub-band and whole band agree within 0.25 dB for targets 200 m apart. For separated targets they differ by up to 2.9 dB, with either sign, but only at levels between -48 and -102 dB. There both schemes are limited by the part of the target's spectrum outside the map block (see Section 2), and the two image sets truncate it slightly differently. At most Doppler-alias collisions (a pair of targets within 0.63 km of a multiple of 6.23 km), the per-sub-band scheme is 1–9 dB *worse* (Fig. 4): its one-sided image sets resolve the overlap of two targets worse.

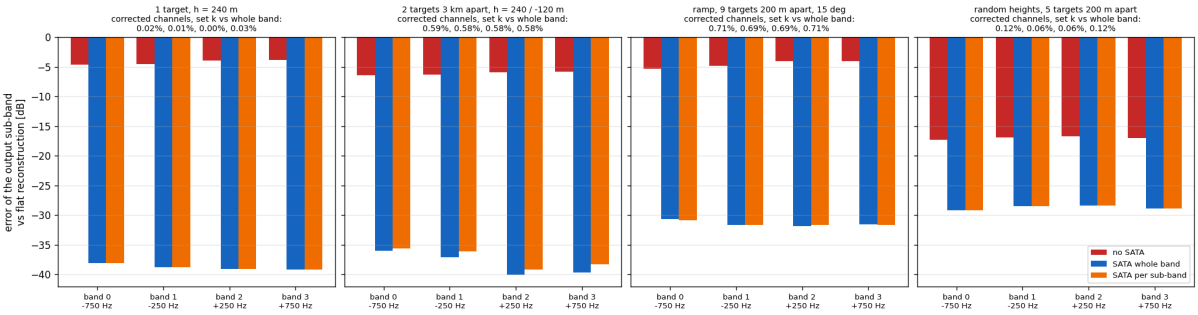


Figure 2: Error of each output sub-band against the flat-terrain reconstruction (lower is better), and the relative difference between the channel set corrected for sub-band k and the whole-band set.

When the per-sub-band scheme can help. Only when the phase a target needs differs from one sub-band to the next by a significant amount:

- **2D processing.** A target's energy in sub-band k sits in a different range bin: at the band edge the range migration is $r_0(1/\cos\beta - 1) \approx 38$ m, several range bins. In 2D the per-sub-band scheme puts the correction where each band's energy actually is.
- **Look-angle-dependent residuals**: tracks that are not parallel (a cross-track baseline that changes within the aperture), squint, very wide beams. Then ΔC_1 and ΔC_2 are not negligible. The per-sub-band map should then carry the full residual phase of the band, $2\pi[\Delta C_0/\lambda + (\Delta D_t - \Delta C_1)f_k + \Delta C_2 \lambda f_k^2]$. The current implementation carries only ΔC_0 per band, which by definition is the same for every band, so as written it would stay equal to the whole band even in those geometries.

In the 1D geometry used here (parallel tracks, constant baselines, broadside), SATA whole band already applies the exact correction to every sub-band, so equal results are the correct outcome.

2 Heatmaps: metric, larger distances

What was wrong. The ambiguity peak was read on $|s|$ in ±20 ms windows around $x_j/v + m\,\Delta t_{amb}$ ($\Delta t_{amb} = 0.81$ s, $m = \pm 1, \pm 2$) and compared with the ideal monostatic signal. That signal is not limited to B_a (about 1340 Hz wide, against 1235 Hz after the reconstruction), so the reconstruction differs from it by the band-edge skirts around every main lobe, even with flat terrain (Fig. 3, top). With targets 2.6 km apart, the windows of one target fall 0.13 s from the main lobe of a neighbour and read that skirt: −43 dB for $N = 5$ with no topography at all. The earlier heatmaps showed this value for SATA in every separated case, and the improvement was capped by it.

New metric: topographic ambiguity. The same reconstruction is run with $b_{xt} = 0$. The targets are iso-range and the receivers then lie on the transmitter track, so the heights leave no phase error: s_{flat} is what the scene would give without topography, skirts and floor included. The topographic ambiguity is the peak of $|s - s_{flat}|$ in the same windows, outside ±20 ms of every main lobe, in dB relative to the peak of s_{flat}. It measures exactly what SATA is meant to remove. Correcting each target with its own ΔC_0 gives −90 to −103 dB, the numerical floor of the metric. The level of $|s_{flat}|$ in the windows is reported separately as the *floor* (−43 to −58 dB); it does not depend on the heights. In the IRF pages the reference curve is now s_{flat}, and the thin black line is $|s - s_{flat}|$.

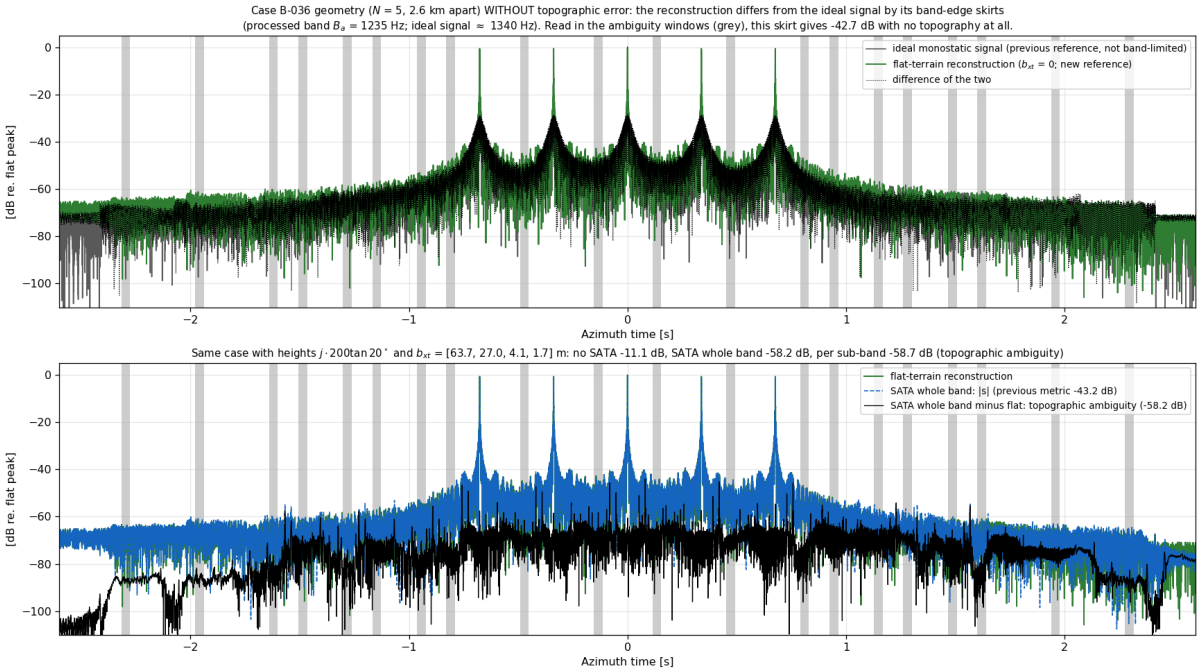


Figure 3: Top: flat terrain, $N = 5$ targets 2.6 km apart. Bottom: the same targets on the ramp $j \cdot 200\tan 20°$.

Configurations. Ramps: heights normal ($h_j = j \cdot 200\tan\alpha$) or large ($h_j = j \cdot 2600\tan\alpha$), targets 200 m, 2.6, 5.2 or 10 km apart (A–H in Table 2), $N \leq 5$ beyond 200 m. Random heights and piecewise ramps: 200 m (all N) and 2.6, 5.2, 10 km ($N = 5$, same heights). The spacings 2.6, 5.2 and 10 km keep every pair of targets at least 1 km from a multiple of 6.23 km. The time axis grows with the scene (±3.1, ±4.1, ±5.1 s), so every target keeps its full aperture and its $m = \pm 2$ windows.

	spacing	heights	cases	no SATA median	SATA whole band median (range)	improvement median (range)	impr., 1°≤α≤5° median (min)	|sub−whole| max	floor range
A	200 m	normal	78	−22.4	−48.0 (−81.8 . . . − 24.2)	24.0 (11.5–33.4)	25.6 (12.0)	0.25	−56 . . . − 48
B	2.6 km	normal	39	−28.3	−76.0 (−101.8 . . . − 53.0)	46.6 (31.6–52.3)	45.7 (45.2)	2.62	−58 . . . − 43
C	200 m	large	78	−7.1	−24.9 (−59.9 . . . − 13.1)	12.6 (5.0–33.0)	20.8 (12.4)	0.21	−56 . . . − 48
D	2.6 km	large	39	−8.6	−59.6 (−95.2 . . . − 51.5)	48.7 (45.3–54.5)	49.9 (46.3)	2.71	−58 . . . − 43
E	5.2 km	normal	39	−28.7	−78.2 (−102.1 . . . − 57.3)	48.5 (32.5–52.6)	48.7 (44.8)	1.72	−46 . . . − 45
F	5.2 km	large	39	−8.5	−58.5 (−92.4 . . . − 51.4)	47.8 (45.0–52.6)	48.9 (46.0)	2.92	−46 . . . − 45
G	10 km	normal	39	−28.4	−75.4 (−102.0 . . . − 53.1)	47.0 (32.6–53.0)	47.0 (44.9)	1.54	−53
H	10 km	large	39	−8.5	−59.0 (−96.8 . . . − 48.9)	48.6 (40.6–53.9)	47.0 (45.1)	2.69	−53
RH	200 m	random	18	−7.7	−17.6 (−33.5 . . . − 11.7)	9.2 (4.8–12.6)	–	0.03	−51 . . . − 48
RH	2.6 km	random	6	−8.9	−58.5 (−69.8 . . . − 51.7)	47.3 (44.2–50.1)	–	2.84	−43
RH	5.2 km	random	6	−8.9	−55.0 (−67.4 . . . − 49.7)	44.8 (42.3–47.6)	–	1.63	−45
RH	10 km	random	6	−9.0	−56.3 (−71.1 . . . − 53.1)	46.9 (45.7–47.6)	–	1.71	−53
PW	200 m	piecewise	15	−13.9	−27.9 (−41.1 . . . − 24.5)	14.6 (7.0–20.3)	–	0.07	−51 . . . − 48
PW	2.6 km	piecewise	5	−16.6	−61.7 (−71.1 . . . − 56.8)	45.0 (39.9–48.6)	–	0.88	−43
PW	5.2 km	piecewise	5	−16.2	−63.9 (−76.5 . . . − 62.3)	47.7 (44.8–49.9)	–	0.55	−45
PW	10 km	piecewise	5	−16.7	−62.9 (−77.4 . . . − 59.1)	46.2 (45.2–47.9)	–	3.19	−53

Table 2: Topographic ambiguity [dB re. flat peak] and improvement [dB]. Floor: $|s_{flat}|$ in the same windows.

- **Separated targets (B, D–H, RH and PW beyond 200 m).** SATA leaves −49 to −102 dB, an improvement of 32 to 55 dB (median 45–49 dB). The level is the same at 2.6, 5.2 and 10 km: beyond one block width ($2h_w = 2.52$ km) the distance no longer matters. In 265 of the 267 separated cases the residual lies below the floor of the flat reconstruction, i.e. it would not be visible in the image. The residual grows with the

height (G: −102 dB at 0.1°, −53 dB at 50°). It is the part of each target's sub-aperture spectrum that falls outside its block of $\pm h_w$ cells and gets no correction. For one target at $h = 3$ km: −62.5 dB with the present h_w, −73.8 dB with $2h_w$, −84.9 dB with $3h_w$.

- **Targets 200 m apart (A, C, RH, PW).** Unchanged in substance: neighbours share one sub-aperture cell ($\delta x = 309$ m) and one correction. The new metric removes the pedestal that hid the small slopes: in A the improvement at $1° \leq \alpha \leq 5°$ is 12.0–33.4 dB (median 25.6).
- **Very small heights.** At $\alpha = 0.1°$ (heights below 5 m) no SATA gives −49 to −70 dB, at or below the floor of the flat reconstruction. SATA still lowers it by 12–41 dB, but the difference would not be visible in the image. These cells are hatched in the improvement heatmaps.
- **Spacing sweep up to 12.5 km** (Fig. 4). The ambiguity after SATA drops with the spacing up to 2.6 km, and beyond that stays at −50 to −64 dB (normal heights) and −49 to −57 dB (large heights), except at 1.6, 2.0, 3.2, 6.2 and 12.5 km, where one pair of targets is within 0.63 km of $6.23\,m$ km (and partly at 4.4 km, where a pair is 0.74 km away). There two targets occupy the same cell of the sub-aperture spectra (same time, same Doppler modulo $\mathrm{PRF}_{\mathrm{ch}}$), and one correction cannot fit both. This limits scenes with regular spacing: no spacing between 2.6 and 16 km keeps 7 or more equally spaced targets clear of it.

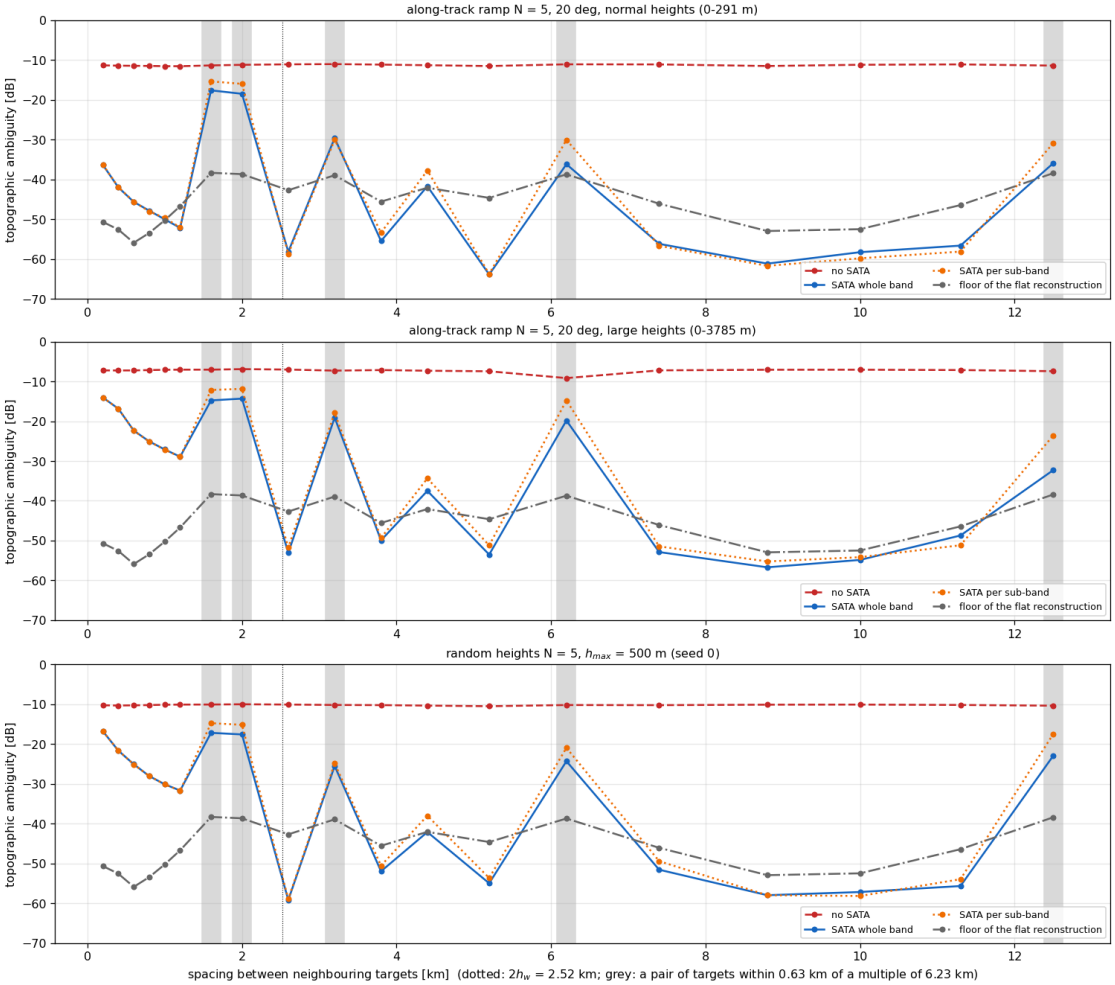


Figure 4: Topographic ambiguity against the spacing between neighbouring targets, same heights, $N = 5$.

The heatmaps of all configurations are in Appendix A (one page for the improvement, one for the levels). The IRF pages of the new spacings are in the companion document.

3 Why the ambiguities saturate at large slopes

3.1 No SATA

Model. With DPCA sampling, channel i samples at $t_n + iT$ ($T = 1/\mathrm{PRF}$). At baseband frequency f_b the channel spectra are $\mathbf{S}_{ch} = \mathbf{A}\,\mathbf{D}\,\mathbf{S}$, where $\mathbf{S}$ holds the N_{rx} folded copies $S(f_b + m\mathrm{PRF}_{\mathrm{ch}})$, $\mathbf{A} = \mathrm{diag}(e^{j2\pi f_b iT})\,\mathbf{F}$ with $F_{im} = e^{j2\pi mi/N_{rx}}$, and $\mathbf{D} = \mathrm{diag}(e^{j\varphi_i})$ holds the phase error each channel has for this target, $\varphi_i = 2\pi\Delta C_{0,i}/\lambda$. The reconstruction applies $\mathbf{A}^{-1}$, so the output is $\mathbf{F}^{-1}\mathbf{D}\mathbf{F}\,\mathbf{S}$, a circulant matrix:

$$S_{rec}(f_b + k\mathrm{PRF}_{\mathrm{ch}}) = \sum_m c(k-m)\,S(f_b + m\mathrm{PRF}_{\mathrm{ch}}), \qquad c(d) = \frac{1}{N_{rx}} \sum_{i=0}^{N_{rx}-1} e^{j\varphi_i}\, e^{-j2\pi di/N_{rx}}. \quad (1)$$

$c(0)$ scales the main peak; $c(d)$ moves the part of the spectrum that survives a shift of $d\,\mathrm{PRF}_{\mathrm{ch}}$ into the image, i.e. an ambiguity at $d\,\Delta t_{amb}$. The processed band is B_a, so only the fraction $\beta_{|d|} = (B_a - |d|\mathrm{PRF}_{\mathrm{ch}})/B_a$ of it survives: $\beta_1 = 0.595$ (-4.5 dB), $\beta_2 = 0.190$. The predicted

ambiguity peak is $|c(d)|\,\beta_{|d|}$. Parseval gives

$$\sum_d |c(d)|^2 = \frac{1}{N_{rx}} \sum_i |e^{j\varphi_i}|^2 = 1. \tag{2}$$

The reconstruction only moves each target's energy between the main peak and its ambiguities; it cannot create energy.

Verification. Fig. 5 sweeps one target from 0 to 6 km. The prediction from the four phases alone matches the simulated ambiguity within 0.3 dB for $m = \pm 1$ and within 1 dB for $m = \pm 2$, where the band edges of the target spectrum are not flat. The largest value in 300 heights is -4.8 dB.

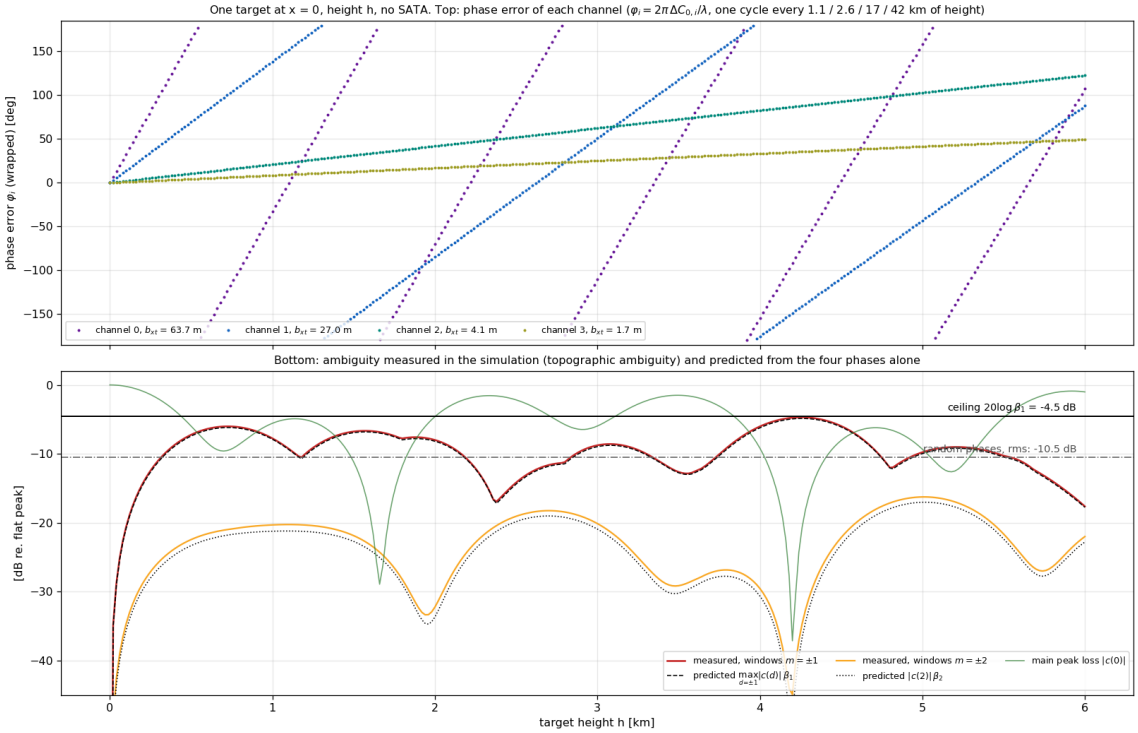


Figure 5: One target, no SATA: channel phases and ambiguity against height.

Why it stops growing. The phase of channel i grows linearly with height, $\varphi_i = 2\pi h/h_{2\pi,i}$ with $h_{2\pi} = 1.1$, 2.6, 17 and 42 km. Two things together cap the ambiguity:

1. *Bound*: $|c(d)| \le 1$, so no ambiguity can exceed $20\log\beta_1 = -4.5$ dB. That value is reached only when all the energy moves into one neighbouring band (φ_i linear in i).
2. *Wrapping*: once the heights span more than one cycle of the long baselines (about 1 km for channel 0, 2.6 km for channel 1), the phases are wrapped and effectively random. Higher targets give different, but not larger, errors. For random phases $\mathrm{E}|c(d)|^2 = 1/N_{rx}$, i.e. -6 dB, or -10.5 dB after β_1.

The heatmap reports the largest value over the targets and the windows, so it sits between the rms and the ceiling, closer to the ceiling the more targets there are. A Monte-Carlo draw of random heights with the four $h_{2\pi}$ reproduces the saturated no-SATA levels of the sweep:

N	Monte Carlo, median	Monte Carlo, 10–90 %	sweep, large heights, $\alpha \geq 20°$, median
1	−9.2	−14.7 . . . − 6.2	–
2	−7.6	−10.6 . . . − 5.7	−7.5
3	−6.9	−9.4 . . . − 5.3	−7.0
5	−6.4	−8.3 . . . − 5.1	−6.7
7	−6.1	−7.5 . . . − 4.9	−6.8
9	−5.9	−7.0 . . . − 4.9	−5.4
13	−5.7	−6.6 . . . − 4.9	−5.3

Table 3: Saturated level without SATA [dB]: largest value over the targets and windows. Largest value in all 456 cases: −4.87 dB (ceiling −4.5 dB).

With normal heights at 200 m the saturation starts later: at 40° for $N = 5$, 30° for $N = 7$, 20° for $N = 9$ and 15° for $N = 13$. In every case the highest target is then at about 650 m, where channel 0 has turned by about 210°. With large heights it starts from $\alpha \approx 3°$. The no-SATA heatmaps therefore look flat at −5 to −9 dB over most of their area. The ambiguity has already reached the level of random channel phases; it does not keep growing.

3.2 SATA, targets 200 m apart

SATA is linear in the data for a fixed map, so each target can be followed on its own: $\mathrm{SATA}(s_{ij}) = g_{ij}\, s_{ij}^{ideal} + (\text{rest})$, with s^{ideal} the exactly corrected channel and g_{ij} its projection gain (ideal: 1). The leakage of target j is then $c_j(d)$ with $e^{j\varphi_i}$ replaced by g_{ij}. This predicts the measured ambiguity after SATA within 0.5 dB in all 13 cases tested (Fig. 6, Table 4).

The mechanism: the bins of target j that point at its own pixel read its own ΔC_0. The bins further out point between j and its neighbours 200 m away and read interpolated values. When the height step Δh between neighbours exceeds $h_{2\pi,i}$, the correction of channel i turns by one or more cycles across the target's footprint. The target's energy is then rotated by a spread of phases, and $|g_i|$ drops to the energy share of the bins nearest the pixel, about 0.1–0.25. This happens for channel 0 at $\Delta h \approx 1$ km and for channel 1 at 1.5–3 km. Channels 2 and 3 ($h_{2\pi} = 17$ and 42 km) stay coherent. With $g \approx (0.23e^{j\psi_0},\, 0.23e^{j\psi_1},\, 1,\, 1)$,

$$|c(1)| = \tfrac{1}{4}\,|0.23e^{j\psi_0} - j\,0.23e^{j\psi_1} - 1 + j| \in [0.24,\, 0.47] \quad \Rightarrow \quad -17 \text{ to } -11 \text{ dB, centre } -13.5 \text{ dB,} \tag{3}$$

which is the plateau of configuration C (−13.1 to −14.2 dB for $\alpha \geq 20°$). It cannot go higher: the two long-baseline channels are already as incoherent as they can get, and the two short ones would only lose coherence for height steps of tens of kilometres within 200 m.

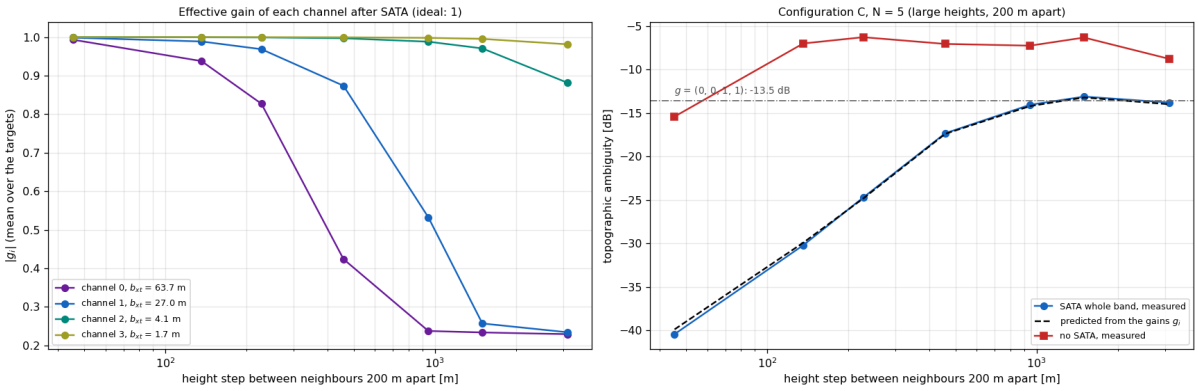


Figure 6: Configuration C, $N = 5$: effective channel gains after SATA, and the ambiguity predicted from them.

heights	N	α	Δh [m]	$\lvert g_0\rvert$	$\lvert g_1\rvert$	$\lvert g_2\rvert$	$\lvert g_3\rvert$	predicted [dB]	measured [dB]
large	5	1	45	0.99	1.00	1.00	1.00	−39.9	−40.4
large	5	3	136	0.94	0.99	1.00	1.00	−29.9	−30.2
large	5	5	227	0.83	0.97	1.00	1.00	−24.8	−24.7
large	5	10	458	0.42	0.87	1.00	1.00	−17.4	−17.3
large	5	20	946	0.24	0.53	0.99	1.00	−14.2	−14.0
large	5	30	1501	0.23	0.26	0.97	1.00	−13.2	−13.1
large	5	50	3099	0.23	0.23	0.88	0.98	−14.0	−13.8
large	2	20	946	0.50	0.77	0.99	1.00	−16.9	−16.7
large	13	20	946	0.09	0.46	0.99	1.00	−14.2	−14.2
normal	5	5	17	1.00	1.00	1.00	1.00	−47.8	−48.2
normal	5	20	73	0.98	1.00	1.00	1.00	−36.0	−36.3
normal	5	50	238	0.81	0.97	1.00	1.00	−24.4	−24.2
normal	13	50	238	0.77	0.96	1.00	1.00	−24.7	−25.2

Table 4: Targets 200 m apart. Δh: height step between neighbours. $|g_i|$: mean over the targets.

With normal heights (configuration A) the steps stay below 240 m, channel 0 keeps $|g_0| \geq 0.77$, and the ambiguity after SATA keeps rising up to 50° (−24 dB) without reaching the plateau. For separated targets each target has its own block, $g_i \approx 1$ on every channel, and there is nothing to saturate.

A Heatmaps

Improvement pages: SATA whole band (top) and per sub-band (bottom); hatched cells are already below the floor of the flat reconstruction without SATA. Level pages: no SATA, SATA whole band, SATA per sub-band.

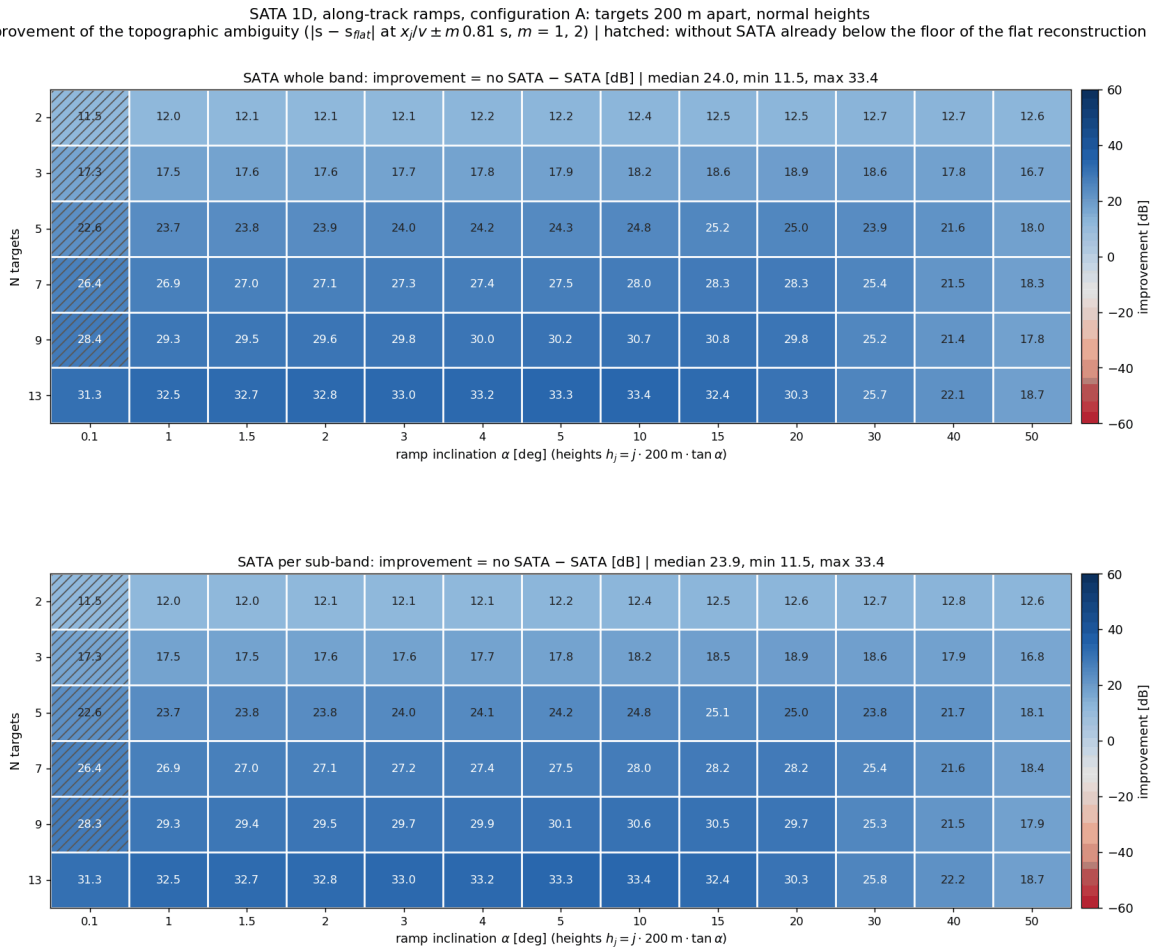


Figure 7: Configuration A: along-track ramps, targets 200 m apart, normal heights: improvement of the topographic ambiguity.

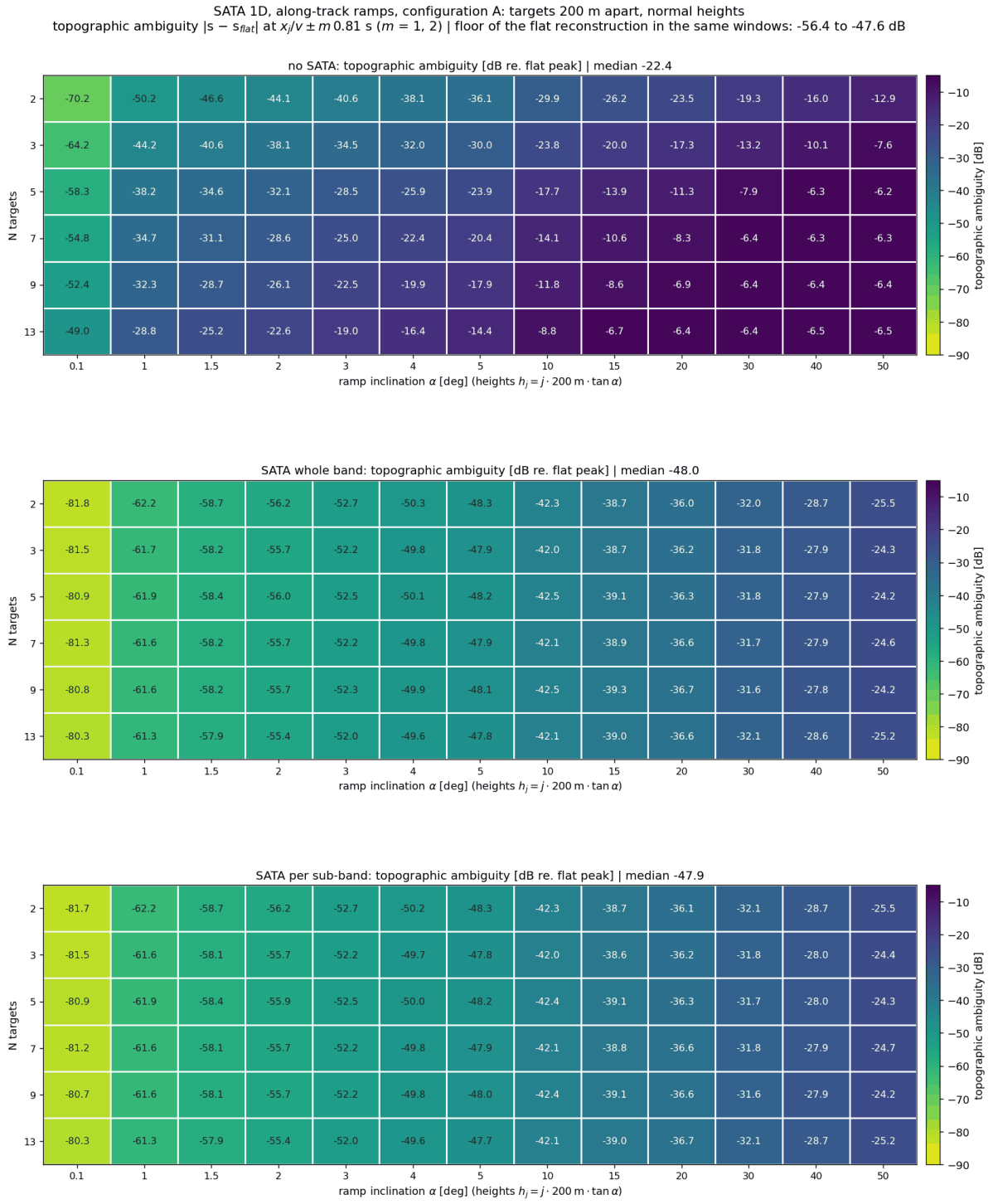


Figure 8: Configuration A: along-track ramps, targets 200 m apart, normal heights: topographic ambiguity.

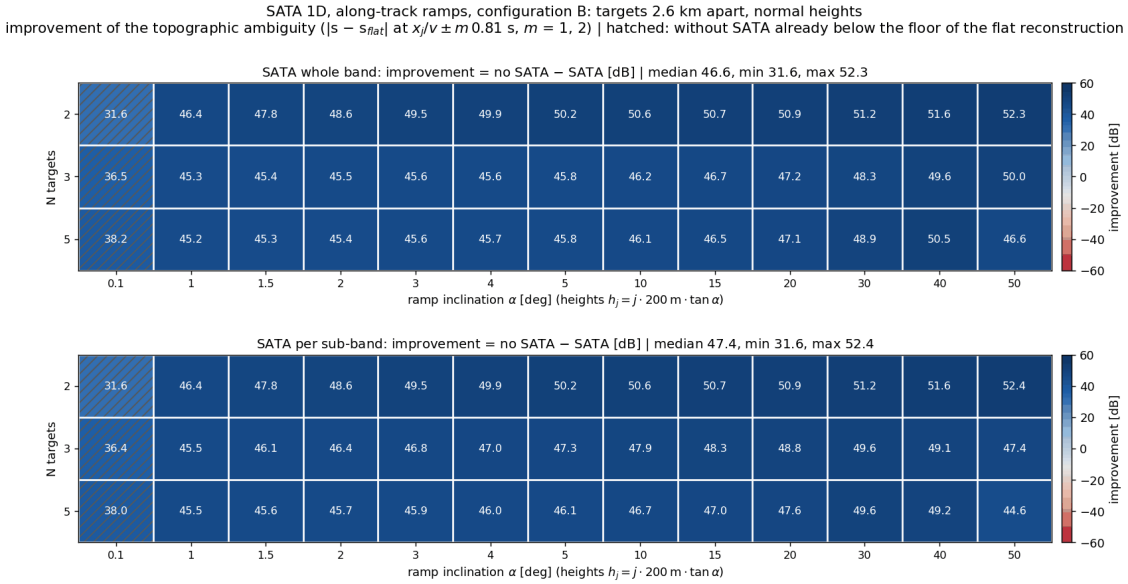


Figure 9: Configuration B: along-track ramps, targets 2.6 km apart, normal heights: improvement of the topographic ambiguity.

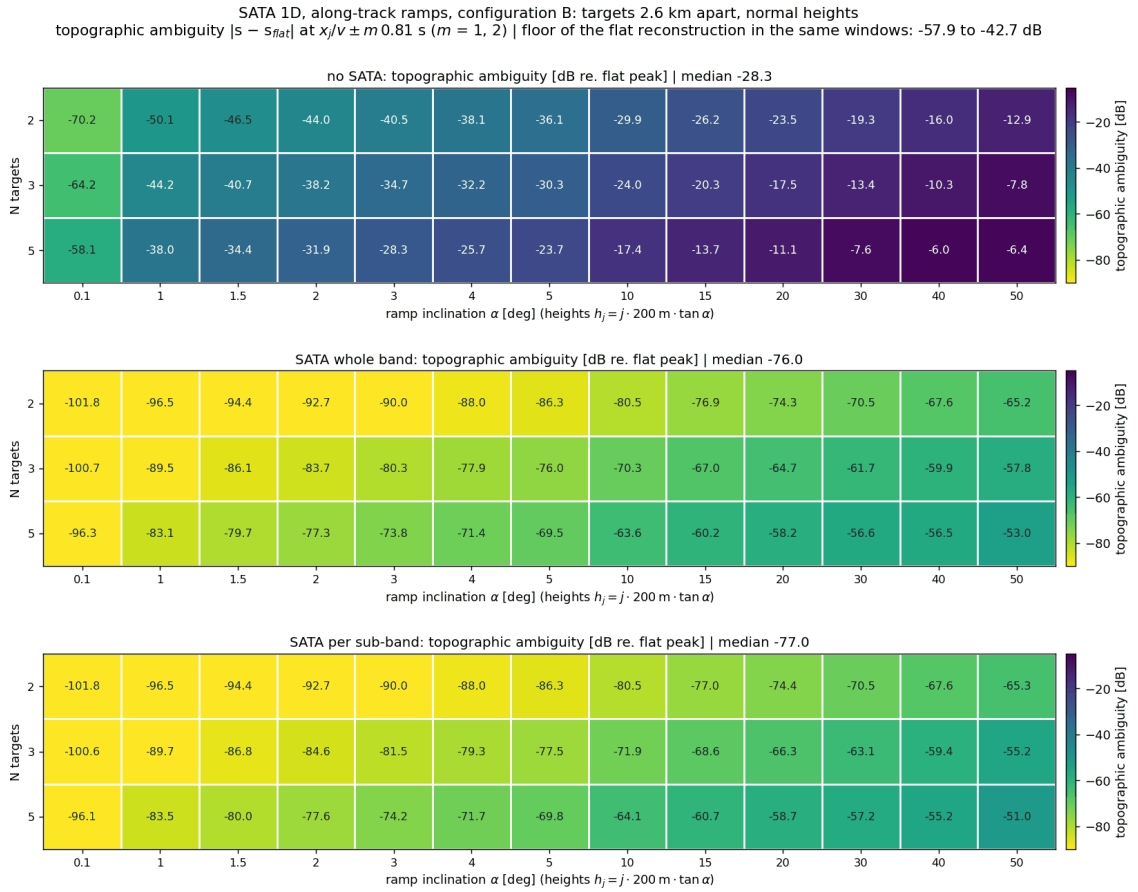


Figure 10: Configuration B: along-track ramps, targets 2.6 km apart, normal heights: topographic ambiguity.

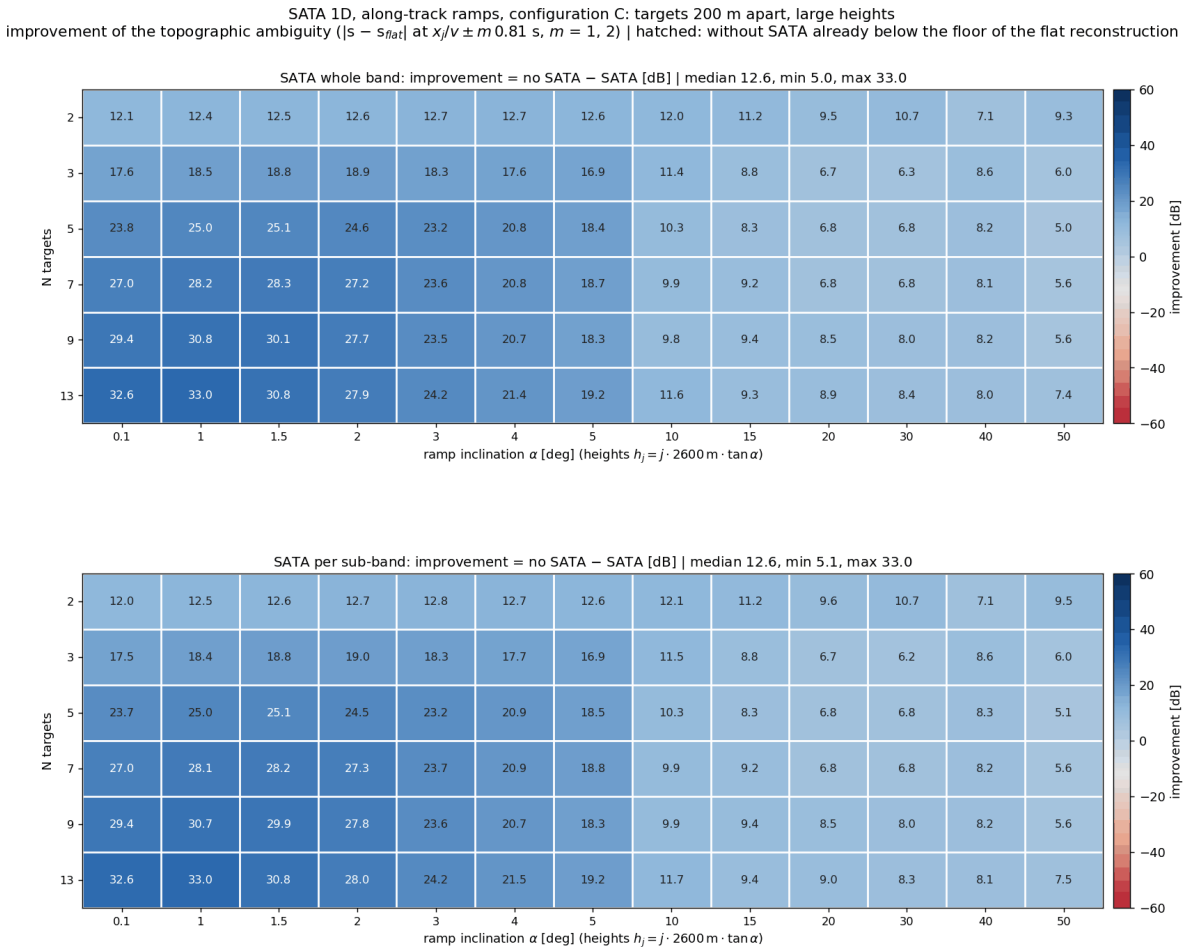


Figure 11: Configuration C: along-track ramps, targets 200 m apart, large heights: improvement of the topographic ambiguity.

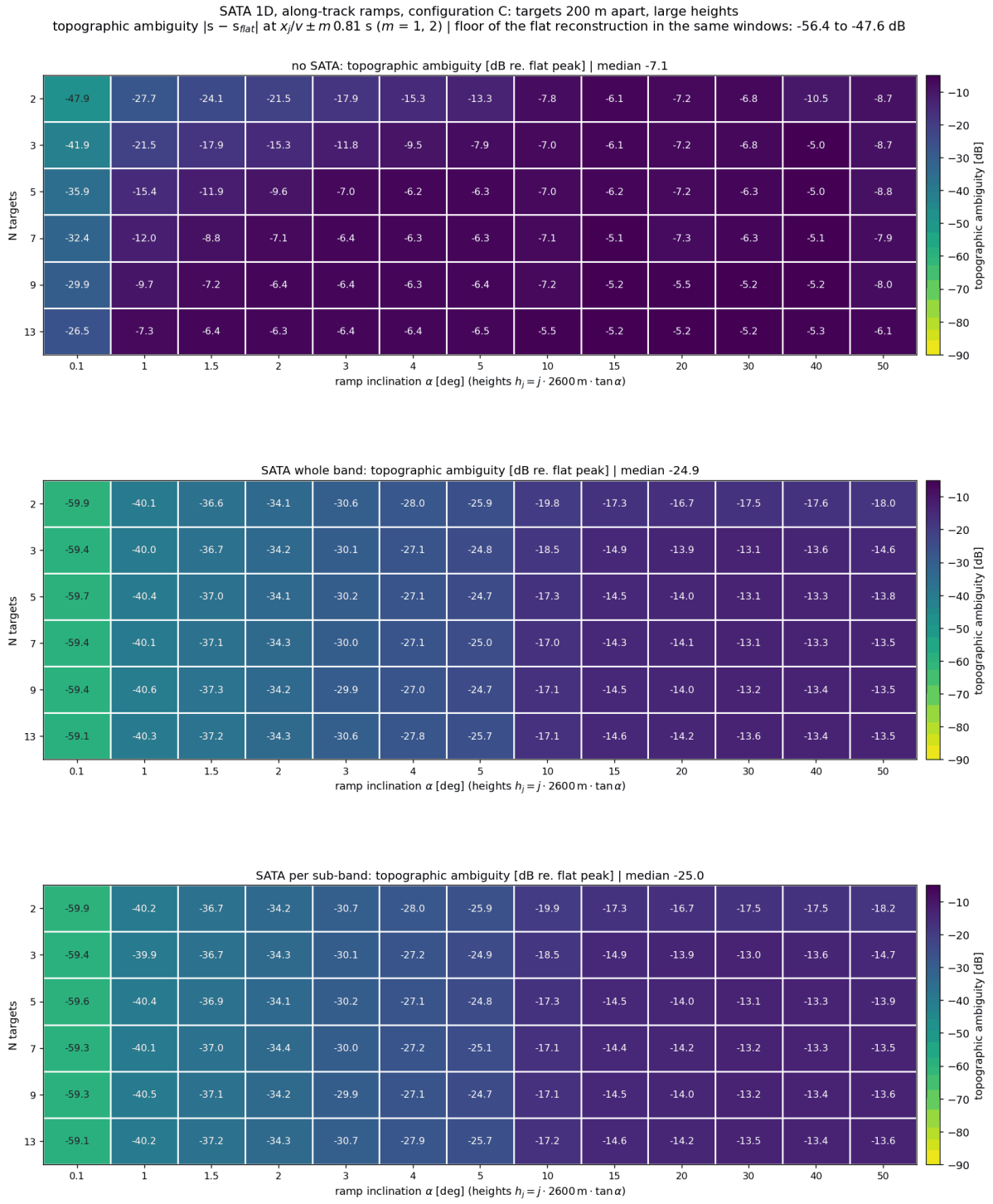


Figure 12: Configuration C: along-track ramps, targets 200 m apart, large heights: topographic ambiguity.

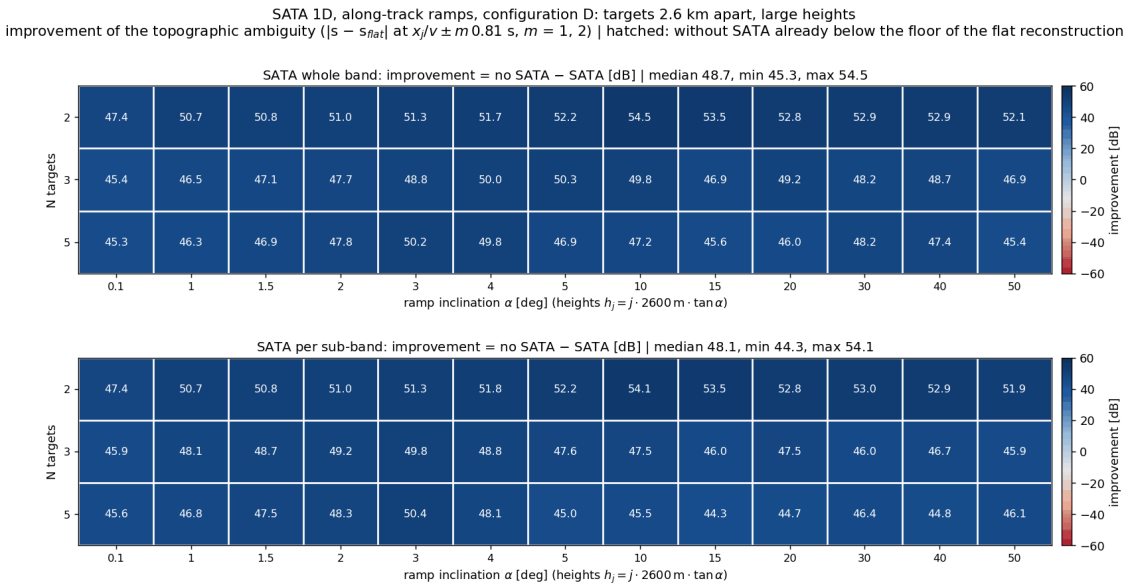


Figure 13: Configuration D: along-track ramps, targets 2.6 km apart, large heights: improvement of the topographic ambiguity.

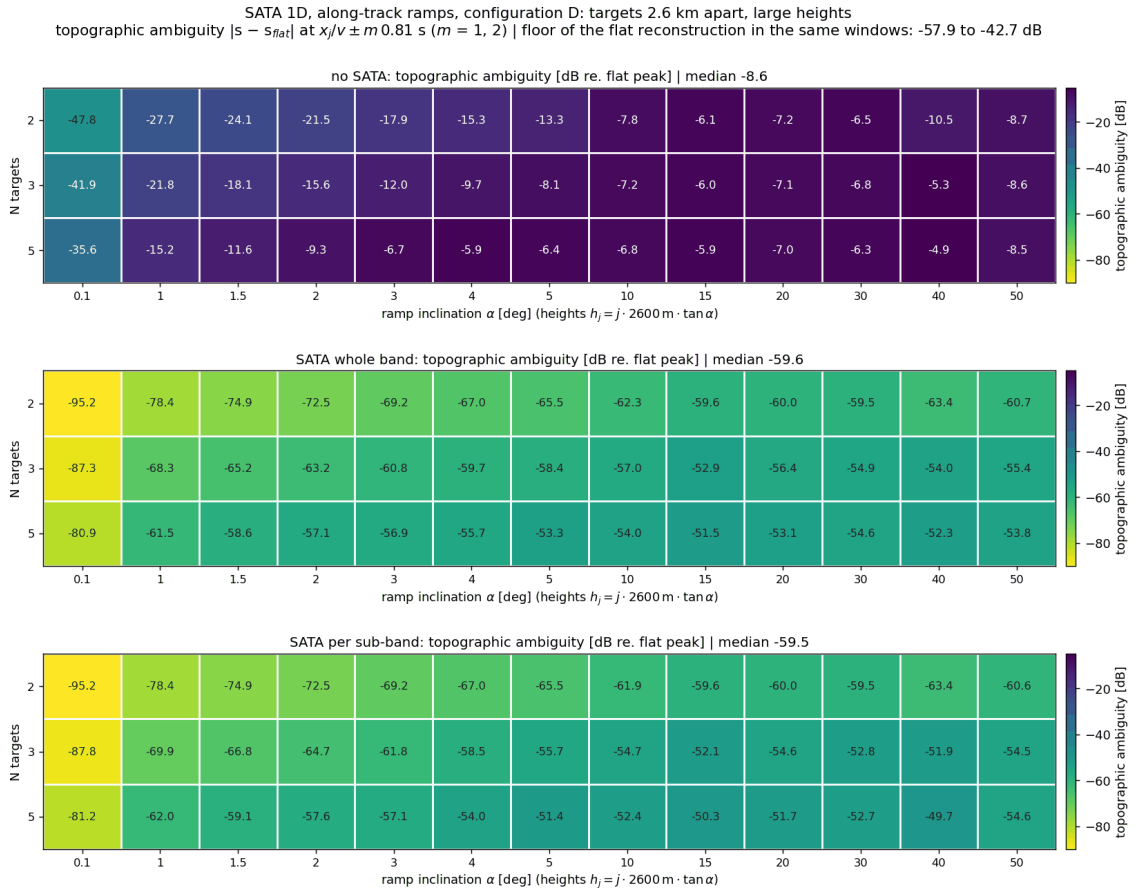


Figure 14: Configuration D: along-track ramps, targets 2.6 km apart, large heights: topographic ambiguity.

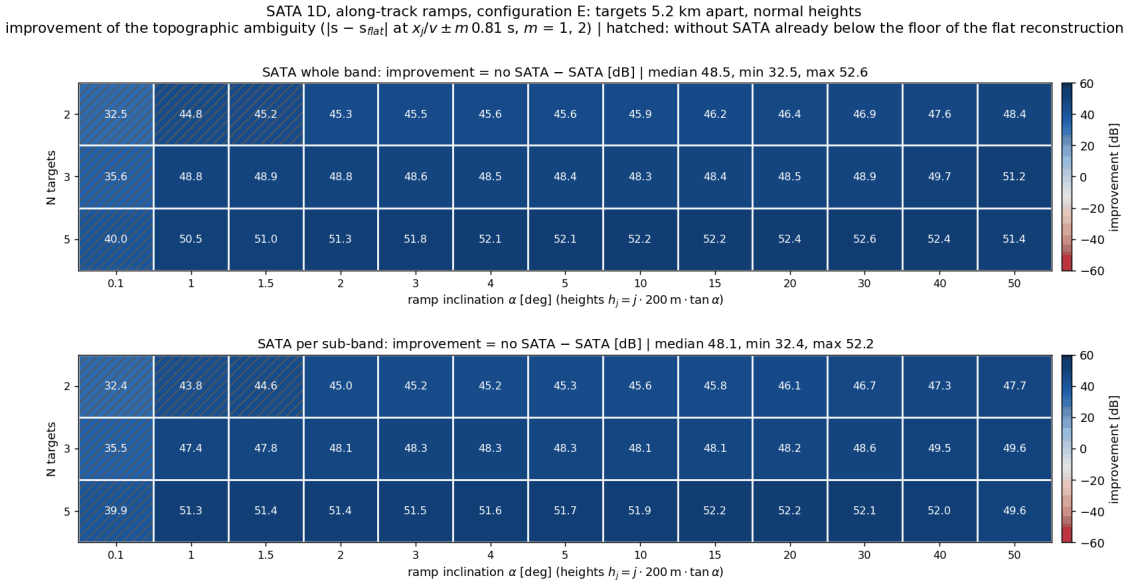


Figure 15: Configuration E: along-track ramps, targets 5.2 km apart, normal heights: improvement of the topographic ambiguity.

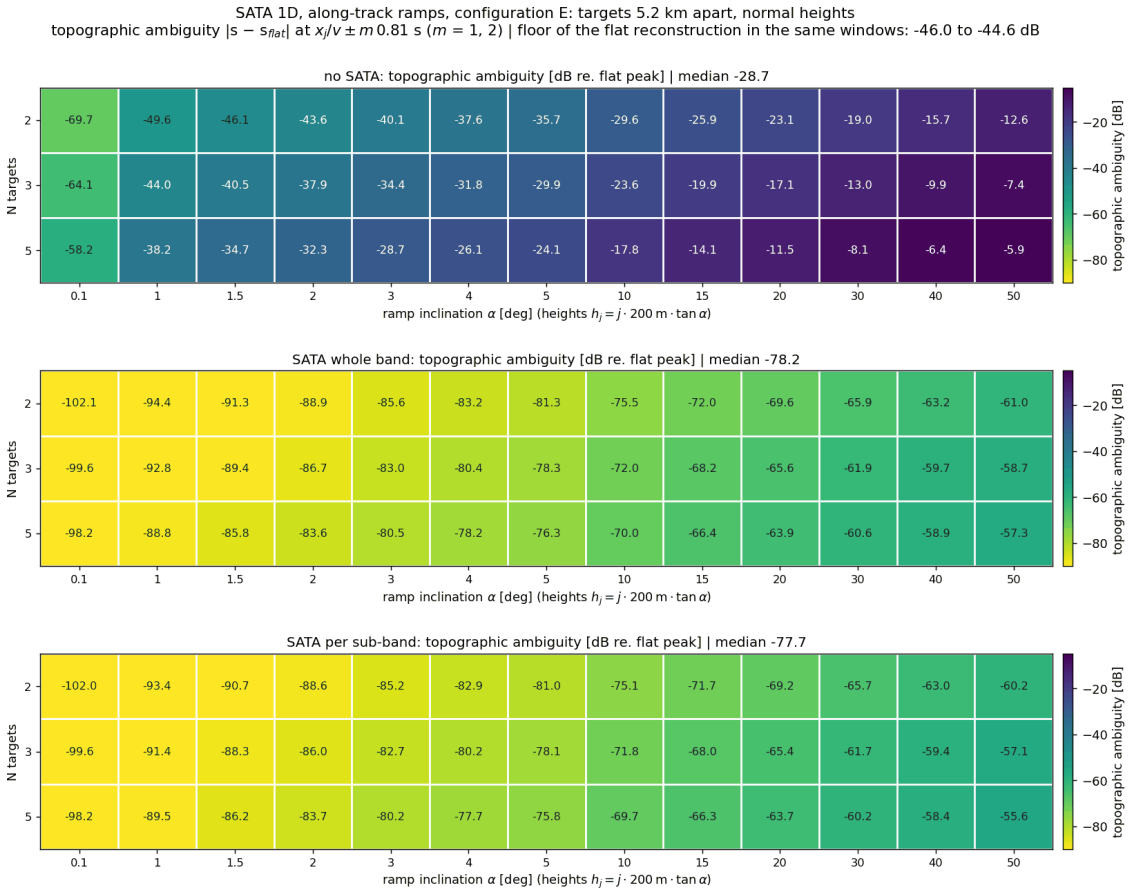


Figure 16: Configuration E: along-track ramps, targets 5.2 km apart, normal heights: topographic ambiguity.

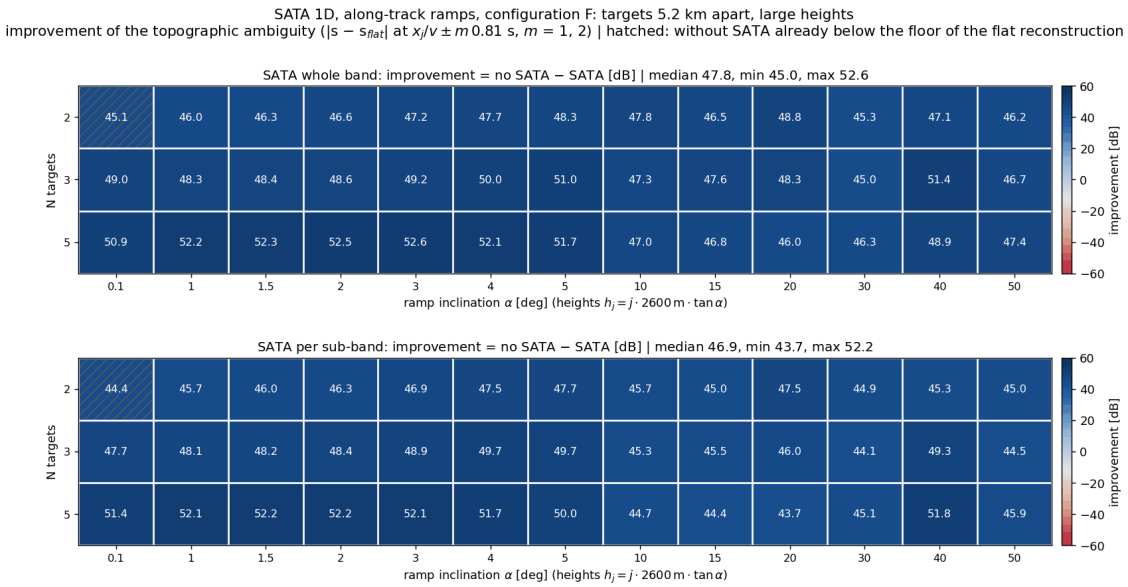


Figure 17: Configuration F: along-track ramps, targets 5.2 km apart, large heights: improvement of the topographic ambiguity.

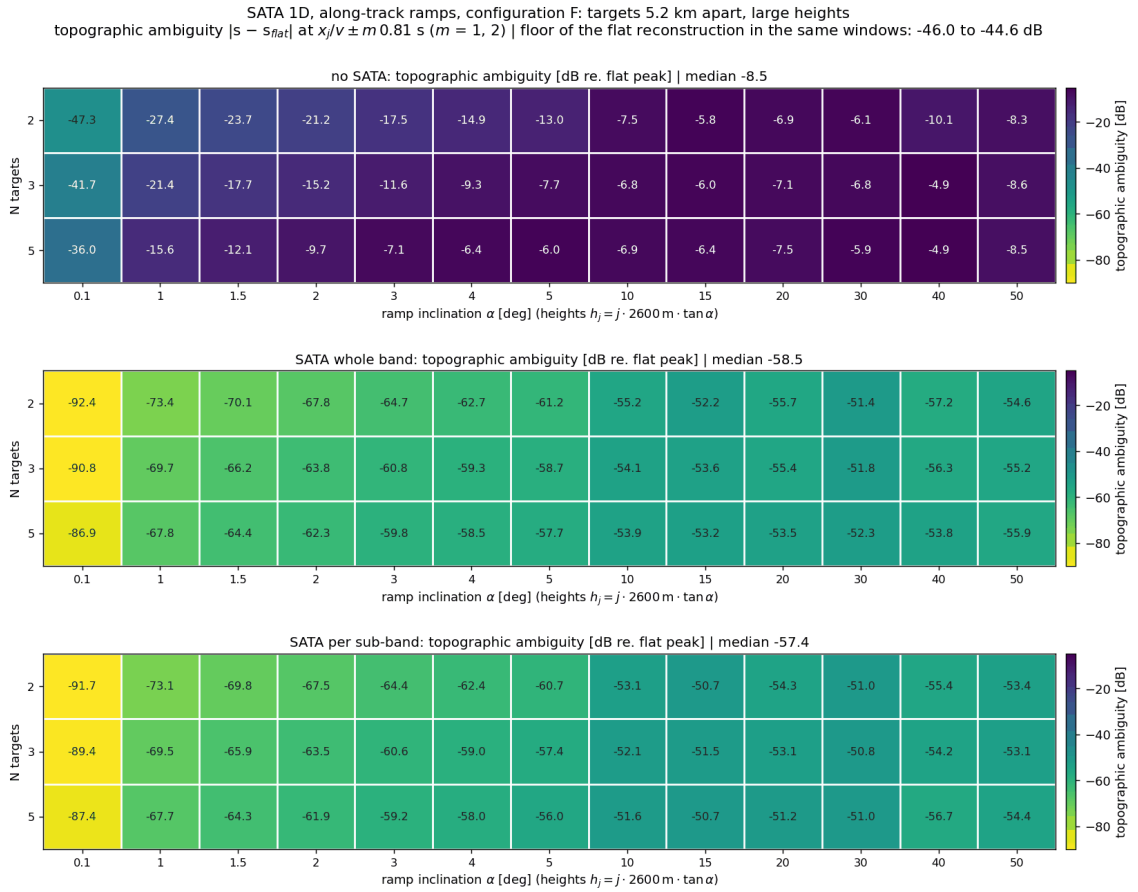


Figure 18: Configuration F: along-track ramps, targets 5.2 km apart, large heights: topographic ambiguity.

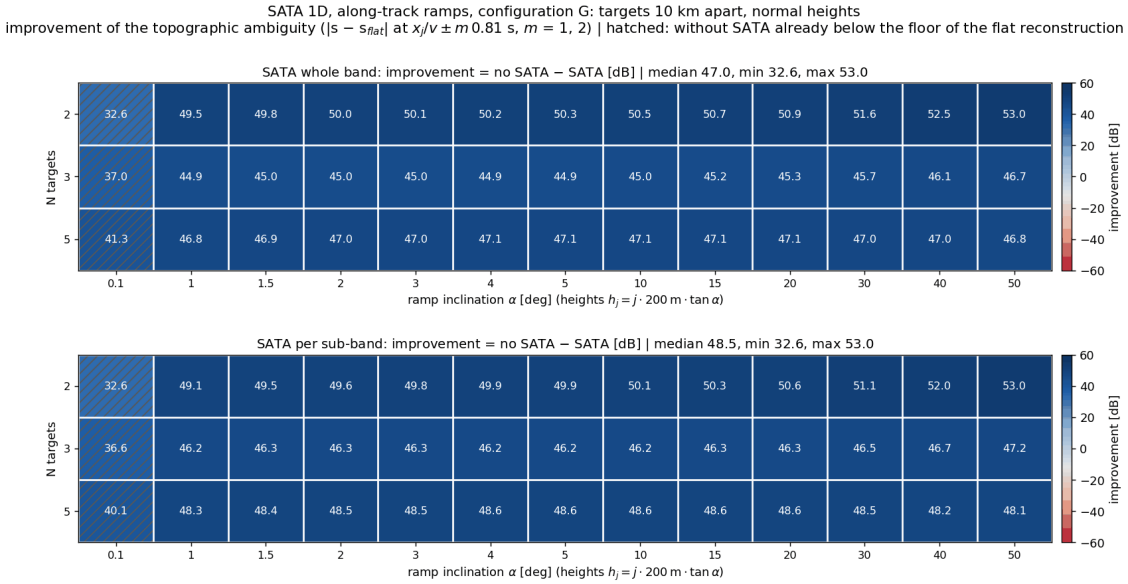


Figure 19: Configuration G: along-track ramps, targets 10 km apart, normal heights: improvement of the topographic ambiguity.

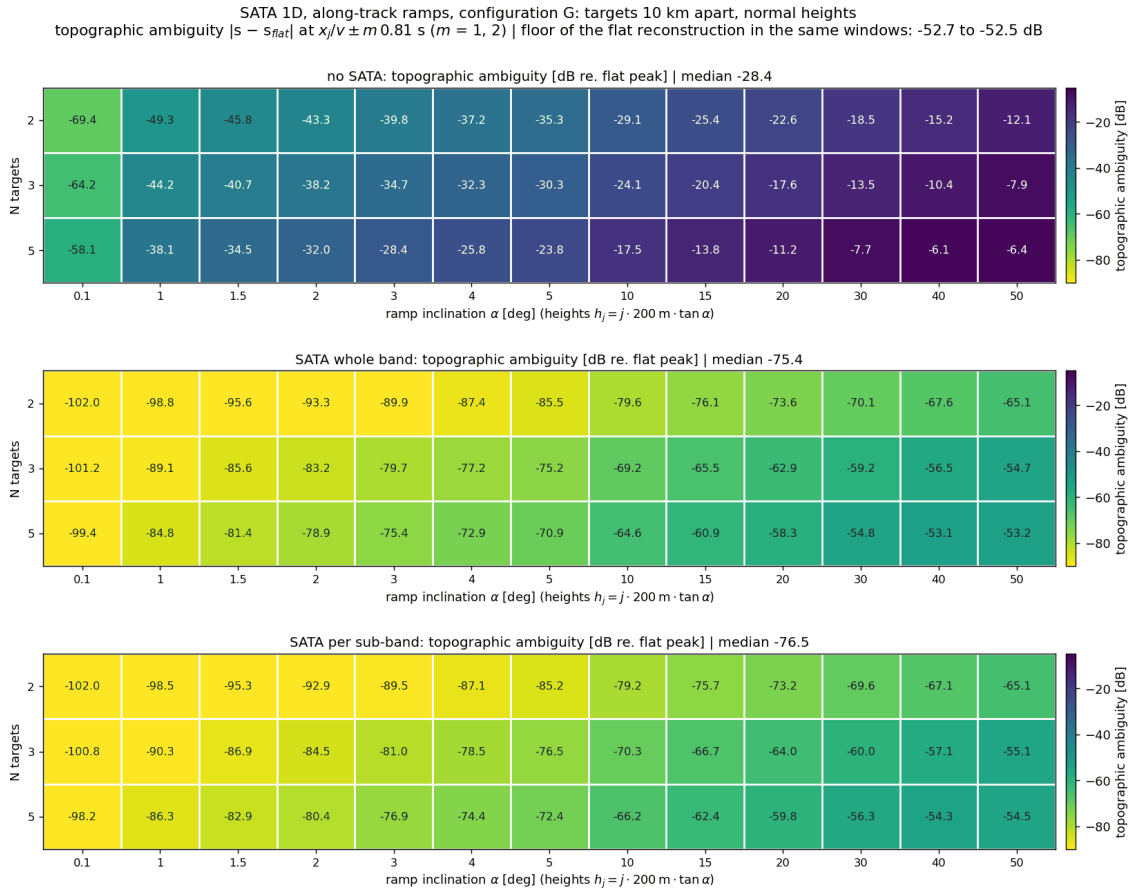


Figure 20: Configuration G: along-track ramps, targets 10 km apart, normal heights: topographic ambiguity.

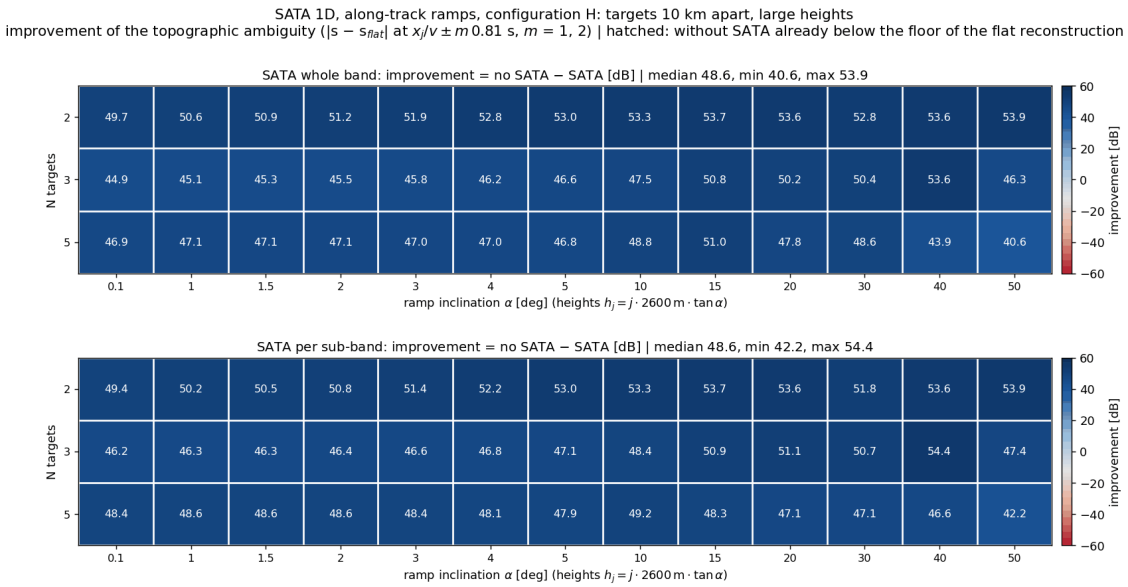


Figure 21: Configuration H: along-track ramps, targets 10 km apart, large heights: improvement of the topographic ambiguity.

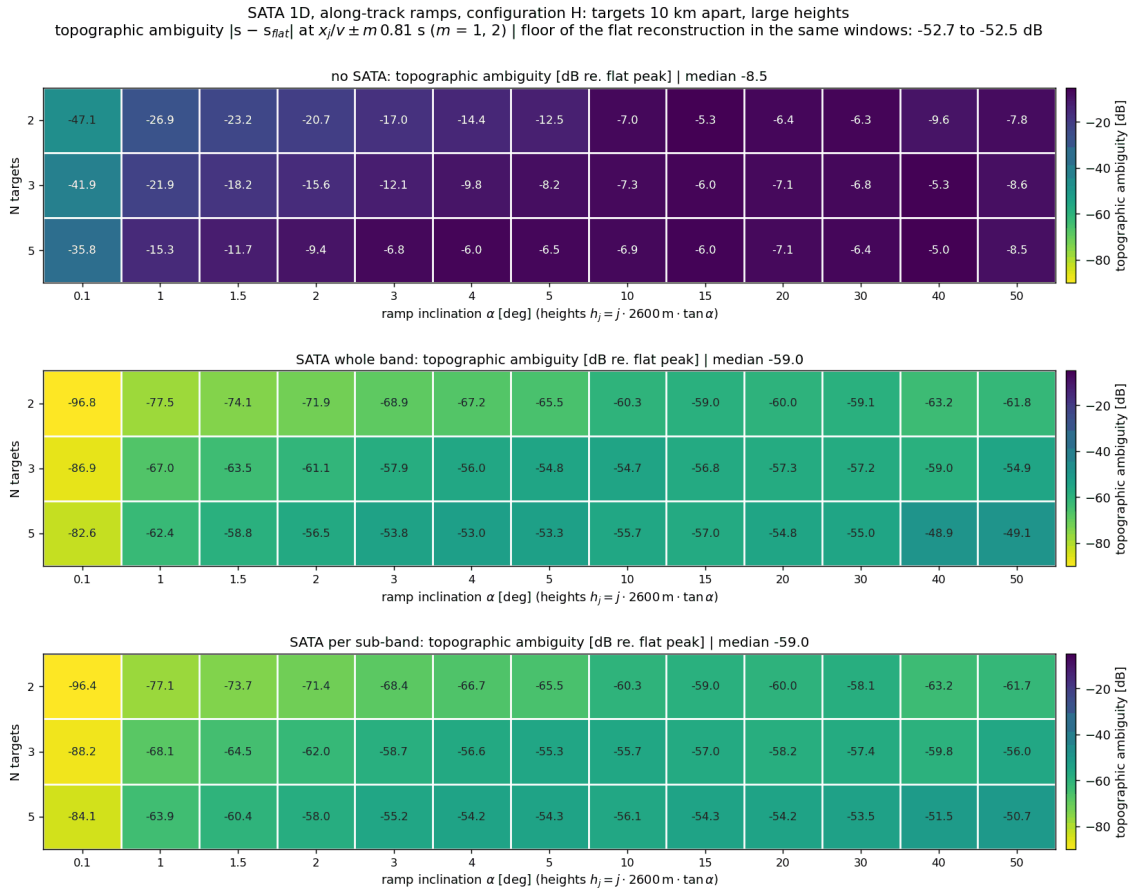


Figure 22: Configuration H: along-track ramps, targets 10 km apart, large heights: topographic ambiguity.

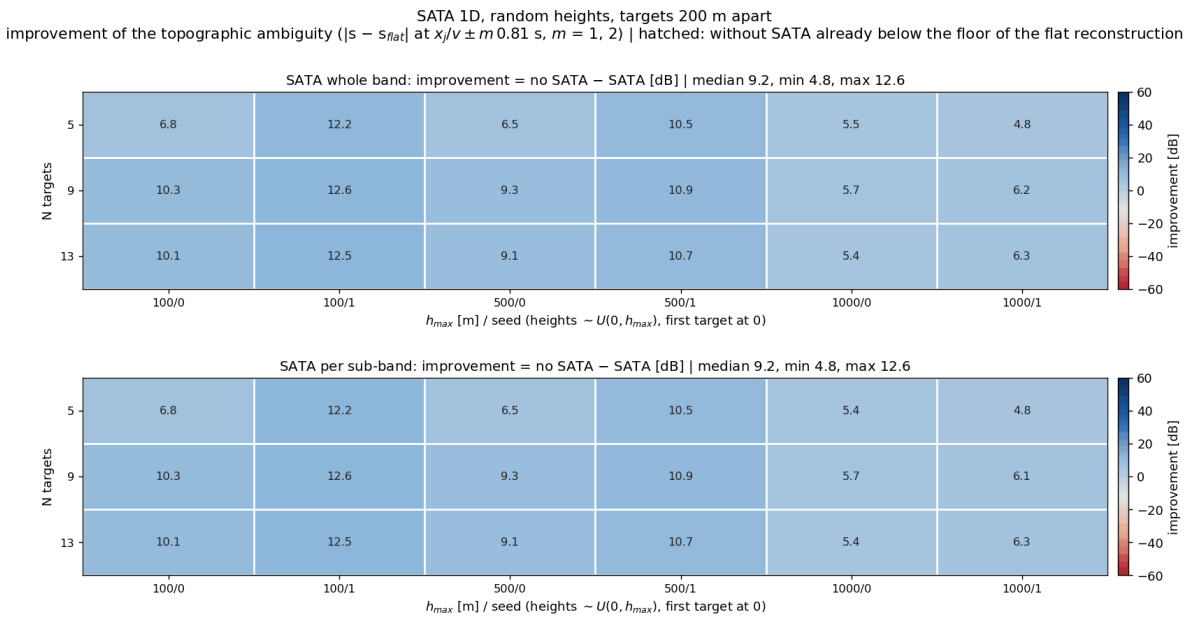


Figure 23: Random heights, targets 200 m apart: improvement of the topographic ambiguity.



Figure 24: Random heights, targets 200 m apart: topographic ambiguity.

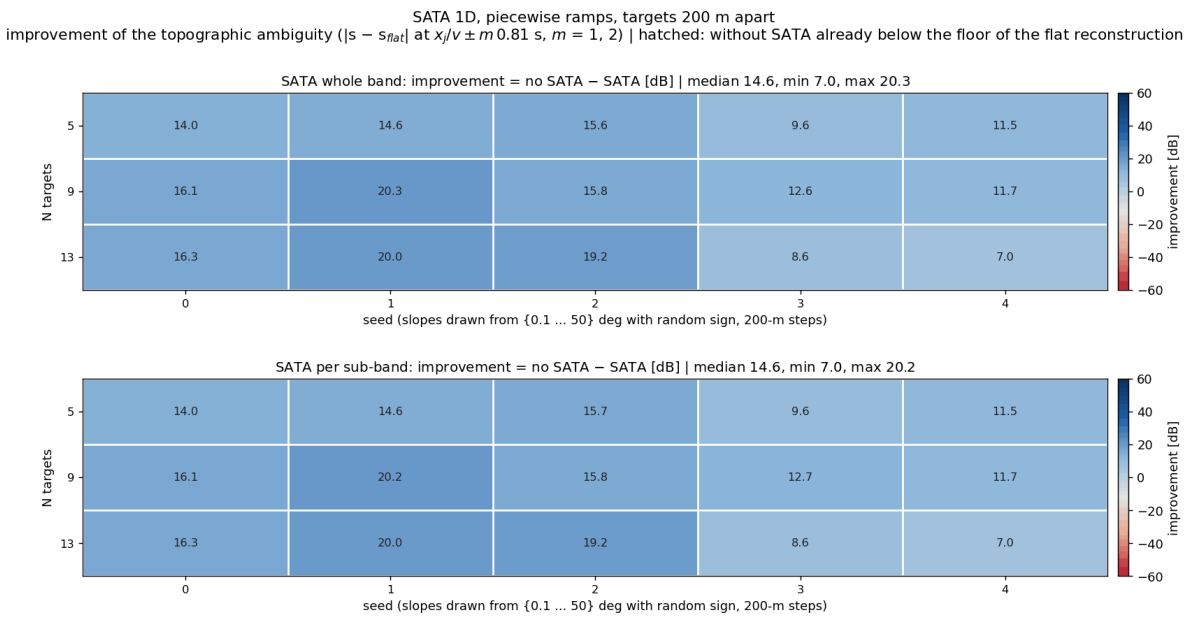


Figure 25: Piecewise ramps, targets 200 m apart: improvement of the topographic ambiguity.

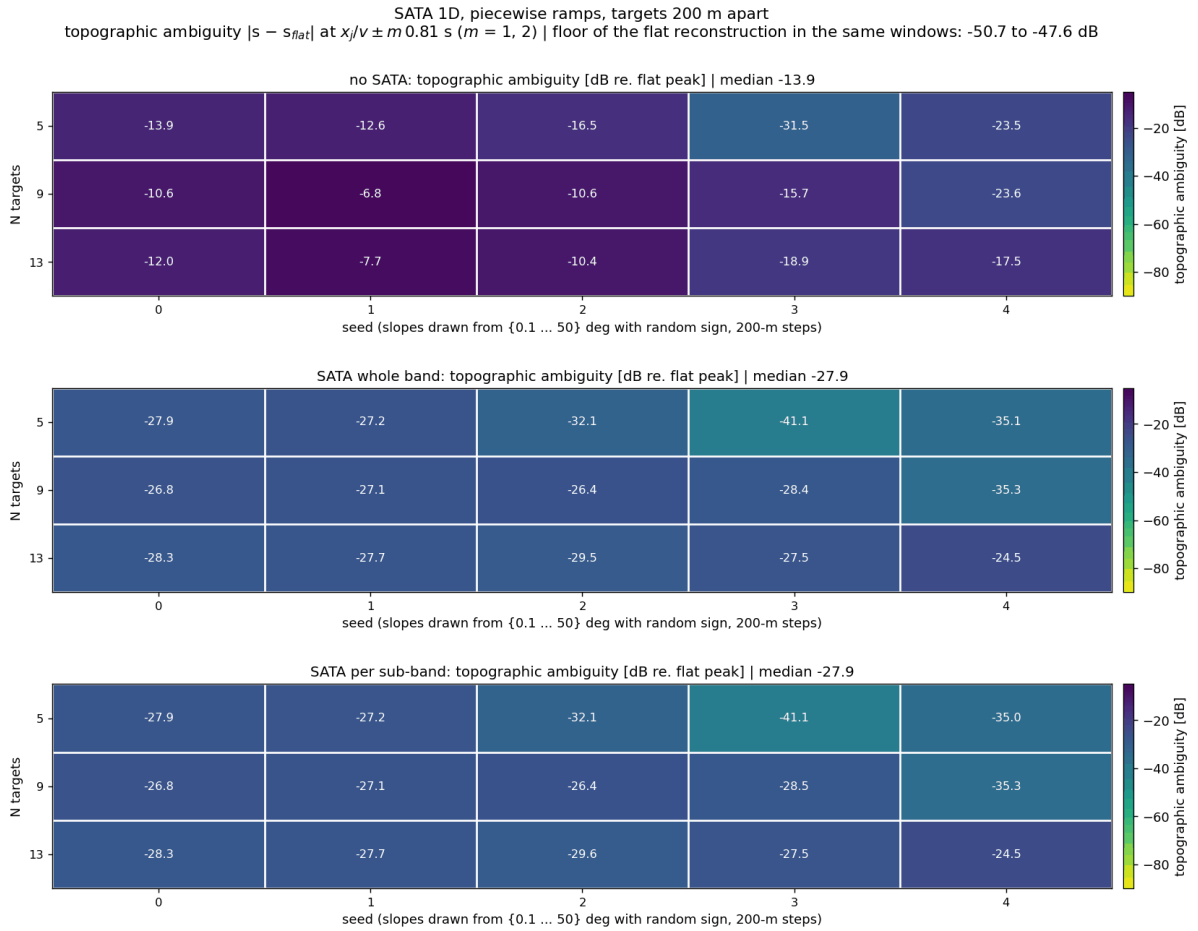


Figure 26: Piecewise ramps, targets 200 m apart: topographic ambiguity.

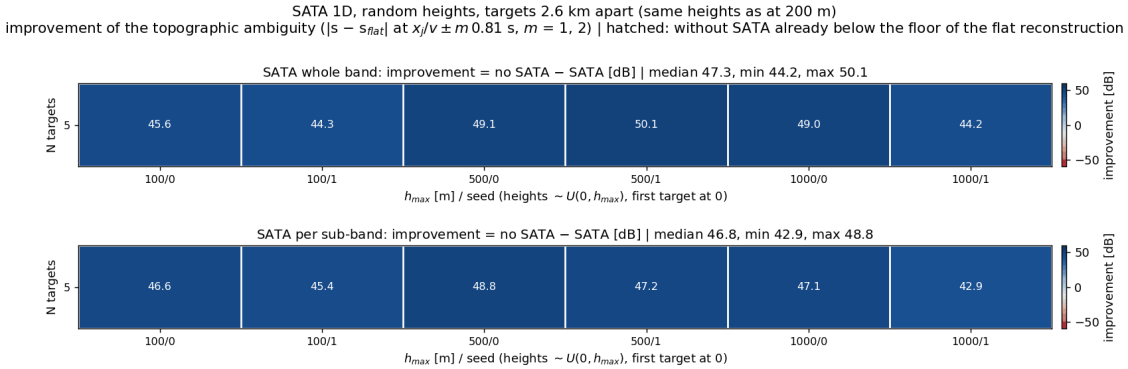


Figure 27: Random heights, targets 2.6 km apart (same heights as at 200 m): improvement of the topographic ambiguity.

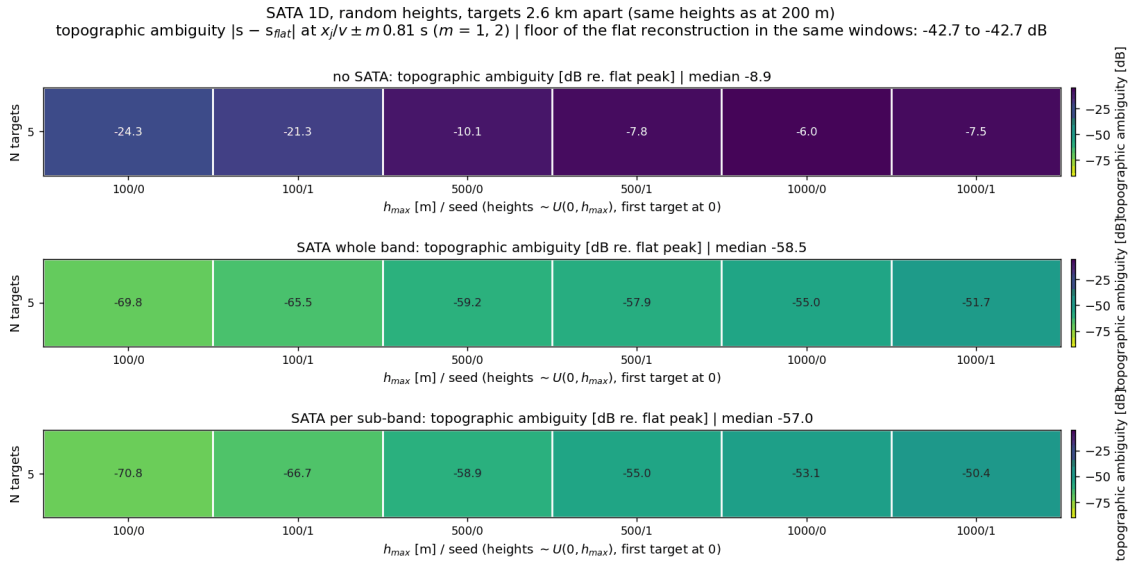


Figure 28: Random heights, targets 2.6 km apart (same heights as at 200 m): topographic ambiguity.

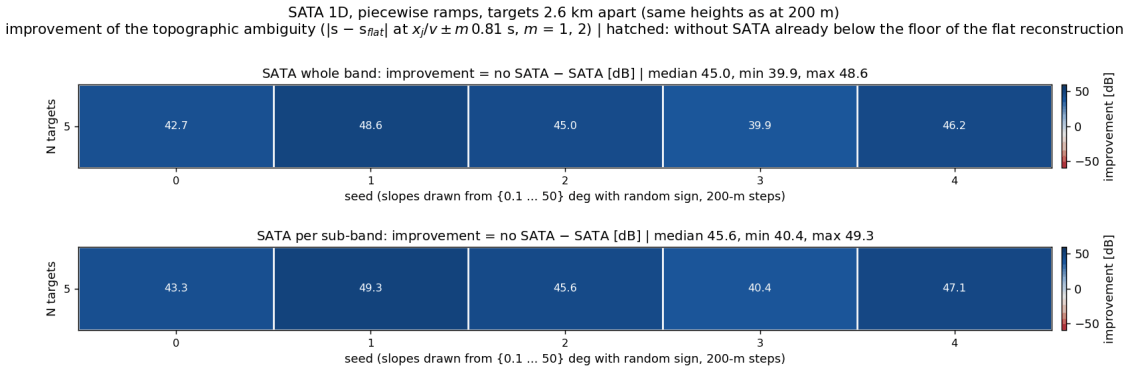


Figure 29: Piecewise ramps, targets 2.6 km apart (same heights as at 200 m): improvement of the topographic ambiguity.

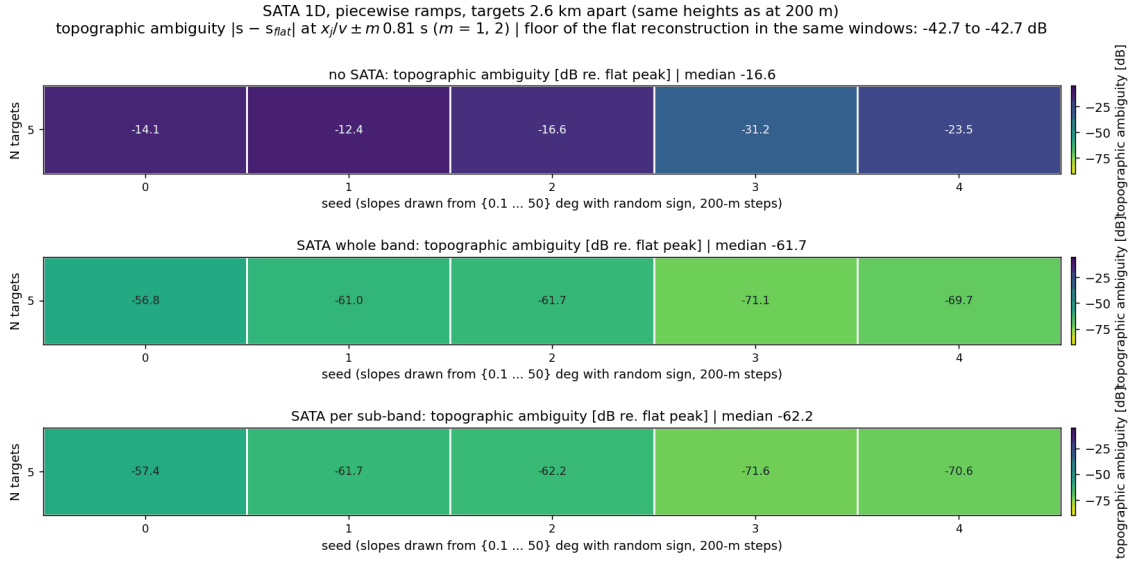


Figure 30: Piecewise ramps, targets 2.6 km apart (same heights as at 200 m): topographic ambiguity.

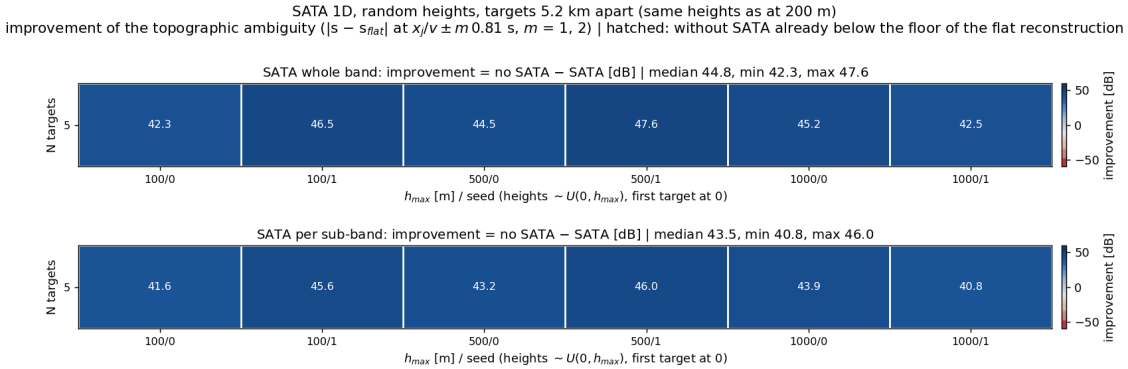


Figure 31: Random heights, targets 5.2 km apart (same heights as at 200 m): improvement of the topographic ambiguity.

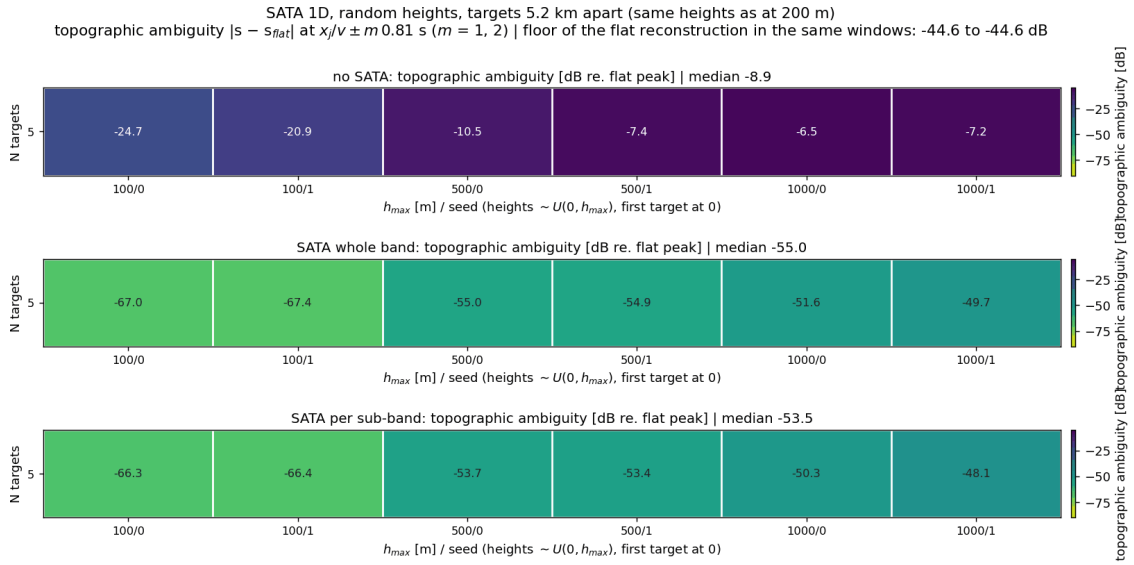


Figure 32: Random heights, targets 5.2 km apart (same heights as at 200 m): topographic ambiguity.

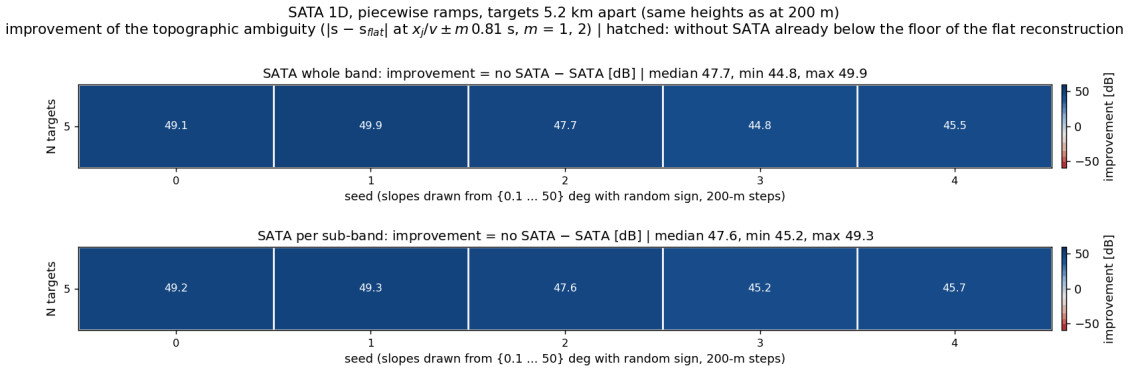


Figure 33: Piecewise ramps, targets 5.2 km apart (same heights as at 200 m): improvement of the topographic ambiguity.

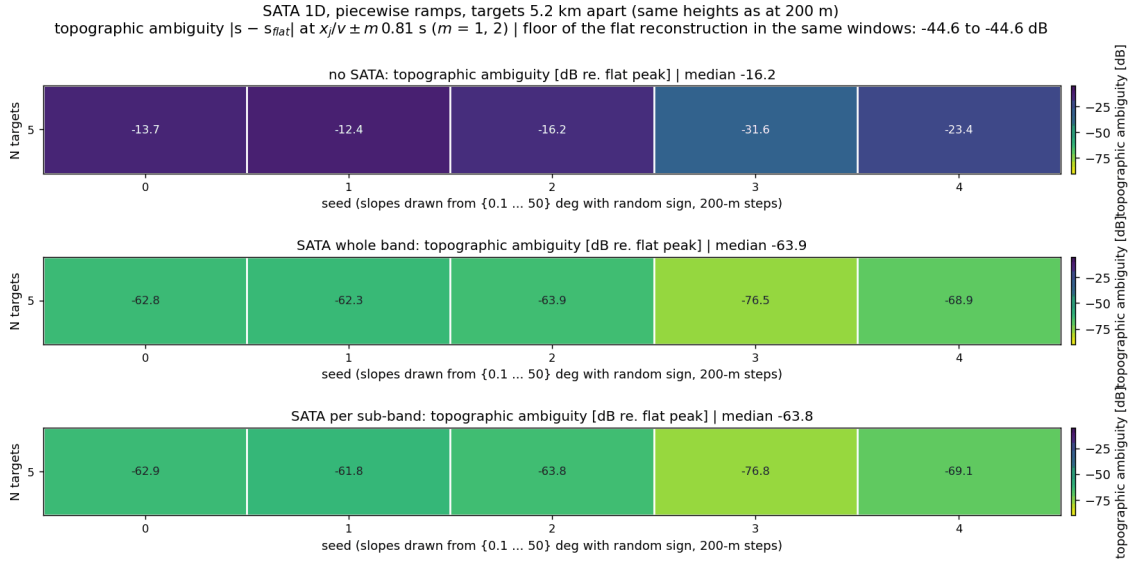


Figure 34: Piecewise ramps, targets 5.2 km apart (same heights as at 200 m): topographic ambiguity.

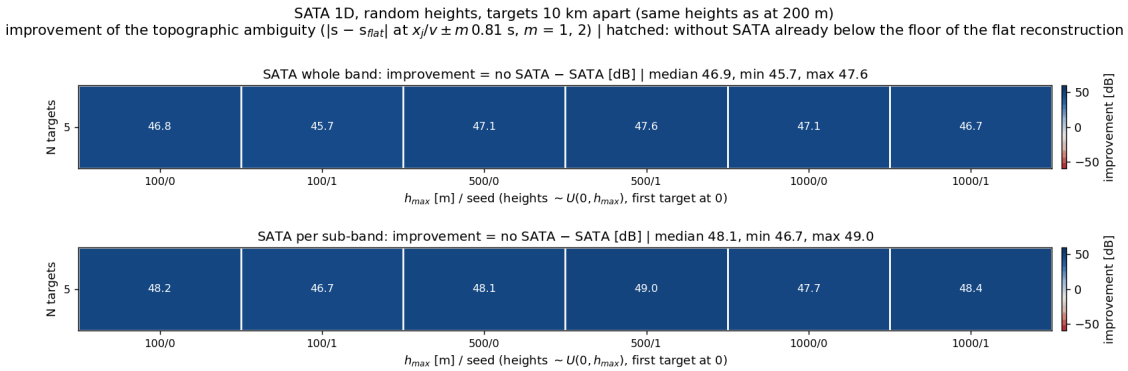


Figure 35: Random heights, targets 10 km apart (same heights as at 200 m): improvement of the topographic ambiguity.

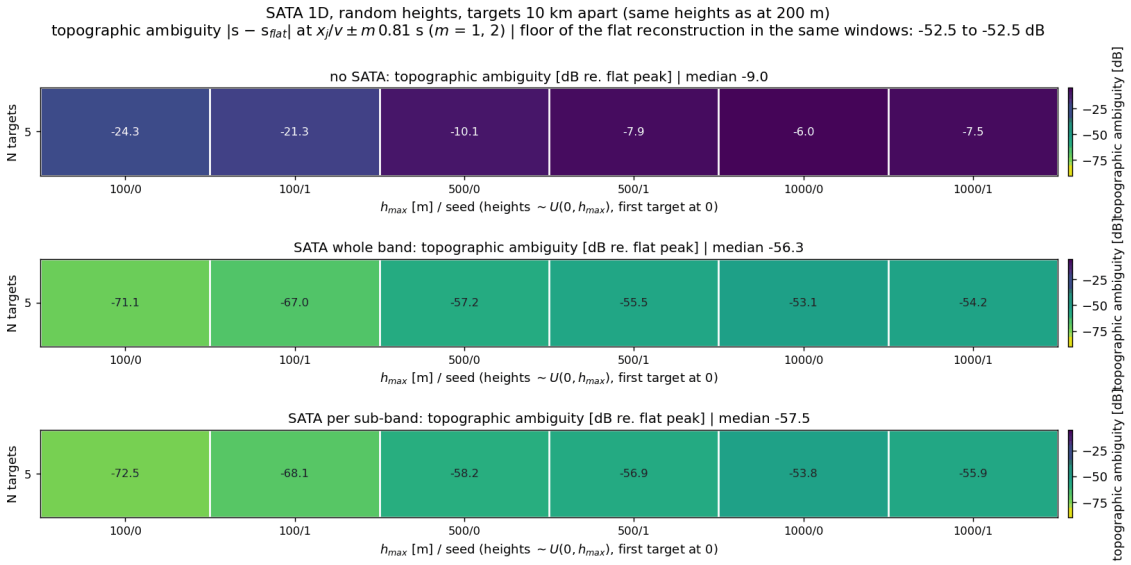


Figure 36: Random heights, targets 10 km apart (same heights as at 200 m): topographic ambiguity.

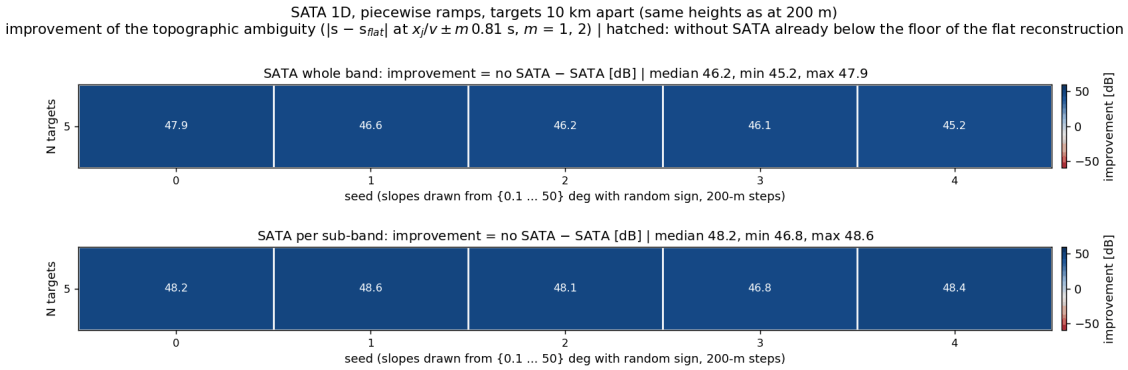


Figure 37: Piecewise ramps, targets 10 km apart (same heights as at 200 m): improvement of the topographic ambiguity.

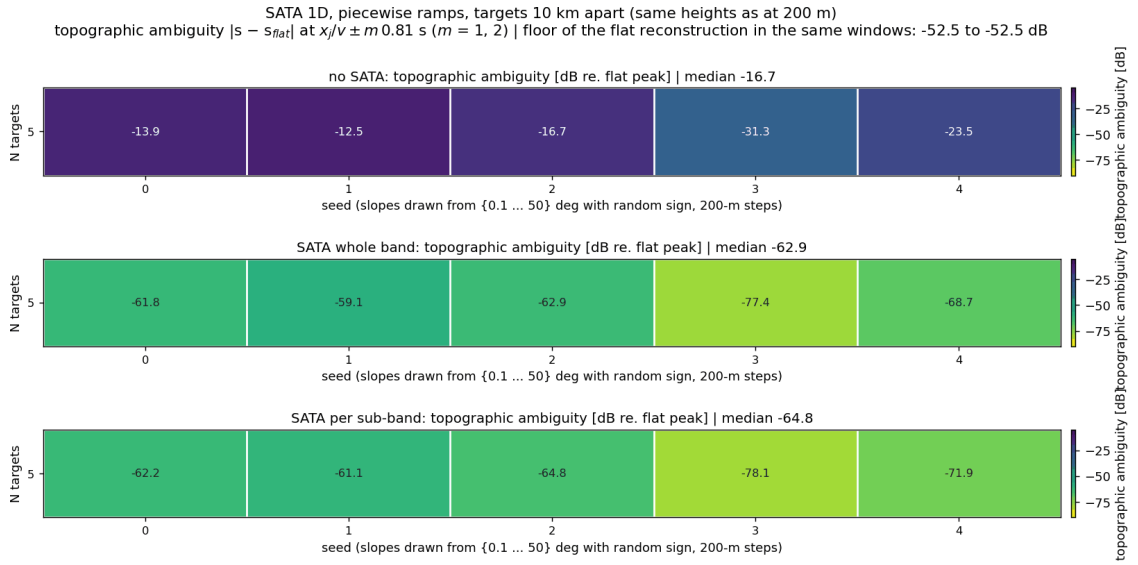


Figure 38: Piecewise ramps, targets 10 km apart (same heights as at 200 m): topographic ambiguity.